

TUNABLE NON-LINEAR MANIFOLD ROMs FOR ELLIPTIC, PARABOLIC, AND HYPERBOLIC PDES VIA MATRIX-FREE PETROV–GALERKIN PROJECTION

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ABSTRACT

Neural operators (Li et al., 2021; Lu et al., 2021) deliver one (accuracy, speed) point per trained model; at deployment only the evaluation grid can change, moving cost, not representation. We present a non-linear manifold reduced order model (NM-ROM) for elliptic, parabolic and hyperbolic PDEs, exposing a family of accuracy/cost points from one decoder, controlled by the correction rank q and three solver-side knobs. The framework combines a matrix-free least-squares Petrov–Galerkin residual projection (Bradbury et al., 2018), empirical-quadrature (EQ) hyper-reduction fitted by non-negative least squares (NNLS) (Hernández et al., 2017; Yano and Patera, 2019), exact boundary enforcement, and a separable decoder: a per-node Fourier-feature coordinate-network bank (Tancik et al., 2020) times a small head with a linear skip, plus nested corrections. Across 2D Burgers, Poisson, heat and waves, the scheduled ladder (one model per rank q) is monotone in q on Burgers on three seeds and a sealed cohort opened once after every choice was frozen (top-rung error 0.59–0.68 % over 4 checkpoints); the fixed-test-count ladder meets a pre-registered bar on one checkpoint; against a fine reference every rung’s error stays within $1.13\times$ the discretisation error; the quadrature rules are confirmed on re-draw at $q \leq 32$ only; and the cost results, each from one job, show the head $2.77\times$ and $6.07\times$ cheaper than the cheapest full-order solve on L-shaped Poisson at 256^2 and 512^2 , with POD-128 cheaper for 15 % more error. Nothing reduced is on the frontier at 256^2 or 512^2 , where a tuned full-order solver is cheaper and more accurate; on Poisson, heat and waves the family collapses to a linear model, showing when the nonlinear manifold pays.

1 INTRODUCTION

Neural operators for partial differential equations (PDEs) (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023; Li et al., 2024; Boullé and Townsend, 2024) have become a strong empirical baseline. But each trained model produces essentially one (accuracy, wall-clock) operating point. The only deployment-time knob is the evaluation grid, and it moves cost without changing what the model can represent (§6.1). Classical Full Order Methods (FOMs) (Hughes, 2000), solved iteratively (Hestenes and Stiefel, 1952), expose this knob through their tolerance. Where a fast transform applies, they are hard to beat on wall-clock.

Two families of methods sit on either side of this gap. Neural operators and PDE foundation models (Herde et al., 2024; McCabe et al., 2024; Hao et al., 2024; Chen et al., 2024) deliver fast amortised inference. But they fix what the model can represent at training time. Reduced order models (ROMs) address the same problem from the opposite direction. Linear projection-based ROMs (Benner et al., 2015; Sirovich, 1987) inherit the FOM’s guarantees. But they hit the Kolmogorov n -width barrier on advection-dominated regimes (Cohen and DeVore, 2015): no fixed linear subspace of modest dimension approximates solutions whose shape moves. Non-linear manifold ROMs (NM-ROMs) (Lee and Carlberg, 2020; Kim et al., 2022) lift this barrier with a learned manifold. But they have rarely been measured head-to-head against neural operators and tuned full-order solvers in the same job. The open question is whether a single trained NM-ROM can expose a deployment-time accuracy/cost tradeoff that neither alternative offers. The second question is where, if anywhere, it is worth having, once the comparators are measured in the same allocation.

This paper answers that question with measurements on 2D Burgers, Poisson, heat and waves. The answer is mixed. We present an NM-ROM framework whose distinguishing property is a *deployment-time accuracy/cost family traced by a single trained model*. At inference, one primary knob and three solver-side knobs trade accuracy for cost. The primary knob is the rank q of a linear correction the solver may switch on. The three solver-side knobs are the iteration cap, the stopping tolerance, and the Empirical Quadrature (EQ) sample count. The machinery is a decoder that enforces Dirichlet conditions exactly, a damped Levenberg–Marquardt latent solver (Marquardt, 1963), Empirical Quadrature (Hernández et al., 2017; Yano and Patera, 2019) validated on reachable states, and a matrix-free JAX implementation (Bradbury et al., 2018).

Contributions.

1. **A tunable NM-ROM with a deployment-time accuracy/cost family from a single trained model, and a matched-dimension result.** With the test count M held fixed so that q is the only control, the Burgers 256^2 ladder (one rung per q) is monotone with every rung converged, spanning $2.44\times$ in same-grid error for $5.16\times$ in cost inside one allocation (§6.3). At matched $k = 16$ inside one bank, the neural head reaches 2.5629 %, where the best linear map reaches 56.9296 % and POD-16 reaches 61.6503 % (§6.4).
2. **Architectural choices that make this family achievable.** We identify three constraints a decoder must satisfy: cold-start convergence, EQ compatibility, and mesh-independent per-node evaluation. We show that a separable decoder (coordinate-network bank, head with a linear skip, nested corrections) meets all three.
3. **Precomputed weak operators and validated quadrature.** Exact preassembly of every linear term, Empirical Quadrature for the one nonlinear term, and a validation rule that accepts a fitted quadrature only by its held-out error on reachable states and by independent re-draws, never by its fitting residual.
4. **Where the family is not worth having, reported as such.** At 256^2 on the square, no reduced subject is on the non-dominated set once a tuned full-order solver and a neural operator on the same data are in the job. On Poisson, heat and waves, the top rung is the linear model and also the cheapest point. On an L-shaped Poisson domain, where no fast transform applies, reduced models are cheaper than the cheapest same-job full-order solve (timed in the same allocation), POD-128 more so than the head (§6.1, §6.5).

What we do not claim. No speedup over an efficient full-order solver on the square; no accuracy superiority over neural operators; no frontier over POD at every rank; no cost ratio across jobs; no theory; and every number is development-cohort evidence, from the cases used while tuning (§6.5).

2 RELATED WORK

Linear and non-linear manifold ROMs. Linear projection-based ROMs (Benner et al., 2015; Sirovich, 1987; Berkooz et al., 1993; Quarteroni et al., 2015; Hesthaven et al., 2016) suffer from the Kolmogorov n -width barrier on advection-dominated and moving-front problems (Cohen and DeVore, 2015; Greif and Urban, 2019; Ohlberger and Rave, 2016). NM-ROMs lift this barrier with a non-linear trial manifold: Lee and Carlberg (2020) introduced convolutional-autoencoder NM-ROMs with LSPG projection (Carlberg et al., 2011), followed by shallow masked autoencoders with sample-mesh hyper-reduction (Kim et al., 2022), reduced over-collocation (Romor et al., 2023), domain-decomposed sparse autoencoders (Diaz et al., 2024), quadratic manifolds (Geelen et al., 2022; Barnett and Farhat, 2022), feature-tracking variants with EQ (Mirhoseini and Zahr, 2023), POD-pre-compressed DL-ROMs (Fresca and Manzoni, 2022) and neural-field ROMs (Kim et al., 2024). We differ from these by a separable decoder — a coordinate-network bank (Tancik et al., 2020) times a small head with a linear skip — augmented by nested *correction* directions chosen offline once, so that a rank q is selected at run time without refit, a family from one trained decoder analogous to runtime-tunable networks (Yu et al., 2019; Cai et al., 2020). NN-augmented projection ROMs (Barnett et al., 2023) and online-adaptive bases (Peherstorfer and Willcox, 2015; Carlberg, 2015) add a nonlinear correction to a linear basis; ours is the reverse composition, the nonlinear head inside the linear bank with the linear part as the correction. The projection is least-squares Petrov–Galerkin (Carlberg et al., 2011) and the linear terms are preassembled as in tensorial POD (Ștefănescu and Sandu, 2014; Weder et al., 2024); we claim no novelty for either.

Neural operators, hyper-reduction, and differentiable programming. Neural-operator surrogates (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023; Li et al., 2023a; Tran et al., 2023; Cao, 2021; Li et al., 2023b; Hao et al., 2023; Wu et al., 2024; Wang and Wang, 2024; Wen et al., 2025; Hu et al., 2025; Shikhman, 2026) and PDE foundation models (Herde et al., 2024; McCabe et al., 2024; Hao et al., 2024; Holzschuh et al., 2025) deliver one fixed (accuracy, speed) point per trained model and typically approximate boundary conditions via penalty terms, with partial fixes via distance functions or hard constraints (Sukumar and Srivastava, 2022; Berrone et al., 2022; Brecht et al., 2023; Wang et al., 2024; Lippe et al., 2023); hybrid solver-in-the-loop methods (Um et al., 2020; Kochkov et al., 2021; Kopaničáková and Karniadakis, 2024; Eshaghi et al., 2025; Raissi et al., 2019; Karniadakis et al., 2021) embed operators in classical iterations. We instead use the network solely for manifold learning with a least-squares Petrov–Galerkin projection (Carlberg et al., 2011) for the residual, and compare against an FNO, a U-Net (Ronneberger et al., 2015; Takamoto et al., 2022) and a Transolver (Wu et al., 2024) on identical data at an equal wall budget (§6.1). For hyper-reduction we retain classical EQ fitted by non-negative least squares (NNLS) (Hernández et al., 2017; Yano and Patera, 2019; Lawson and Hanson, 1974) over DEIM (Chaturantabut and Sorensen, 2010), ECSW (Farhat et al., 2014; Chapman et al., 2017), GNAT (Carlberg et al., 2013) and their neural and high-order variants (Negri et al., 2015; Hirsch et al., 2024; Nguyen, 2024); the difference is that only the nonlinear term is sampled and that a fitted rule is accepted by its held-out error and by re-draws, not by its fit residual. On separable cells the discrete sine transform (Swarztrauber, 1977) is a direct solve faster than every reduced model we measure; preconditioned Krylov methods (Hestenes and Stiefel, 1952; Saad, 2003) are the iterative comparator.

3 METHODOLOGY

We turn a learned manifold into a practical PDE solver through three components: a trial manifold whose decoder enforces Dirichlet conditions exactly and carries nested linear correction directions (§3.1); a least-squares Petrov–Galerkin projection of the discrete residual onto fixed weak tests (§3.2); and exact preassembly of every linear term with Empirical Quadrature for the one nonlinear term (§3.3); §3.4 gives the architecture. Throughout, $u \in \mathbb{R}^n$ is the full-order state on a uniform grid of N intervals per axis, $A \in \mathbb{R}^{n \times n}$ the negative five-point Laplacian on $[0, 1]^2$; R is the bank width, k the latent dimension, M the number of weak tests and m the number of quadrature nodes. Figure D.1 (Appendix D) shows the data flow; Appendix B gives the per-PDE derivations and exit codes.

3.1 TRIAL MANIFOLD WITH EXACT DIRICHLET ENFORCEMENT

We train a decoder $\mathcal{D} : \mathbb{R}^k \rightarrow \mathbb{R}^n$, $k \ll n$, with latent state z and no encoder in the deployed path, separable into a frozen spatial *bank* $G \in \mathbb{R}^{n \times R}$ times a small neural *head* $h_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^R$, $k \leq R \ll n$, the bank built once per mesh from a random-Fourier-feature coordinate network g_ϕ (Tancik et al., 2020),

$$G_{x,:} = \mu(x) g_\phi(x)^\top, \quad \mu(x) = 16 x_1(1 - x_1) x_2(1 - x_2). \quad (1)$$

To strictly enforce the homogeneous Dirichlet condition on the boundary Γ , we utilize the smooth vanishing factor μ , zero on Γ , folded into every column of the bank, so that $\partial u / \partial z = 0$ on Γ at every resolution (homogeneous data only).

The correction ladder. The deployed model augments the head’s output with q fixed directions $C_q \in \mathbb{R}^{R \times q}$ and solves for the latent code and the correction coefficients together,

$$u(z, y) = G(h_\theta(z) + C_q y), \quad z \in \mathbb{R}^k, y \in \mathbb{R}^q, 0 \leq q \leq R. \quad (2)$$

The columns of C_q are chosen offline once per checkpoint by a principal component analysis of the head’s own residual on the training codes, ordered so that C_q is a prefix of $C_{q'}$ for $q < q'$; the ladder is nested and no rung needs retraining. At $q = 0$ the model is the frozen head; at $q = R$ the head is irrelevant and the model is the linear reduced model on the bank’s span, solved without a nonlinear iteration; in between, q moves the reachable set from the head’s image towards the bank’s span, and the solved dimension from k to $k + q$. Two consequences follow: the bank’s projection error is a floor on the achievable error whatever the head or the solver does, so nonlinearity in h_θ buys a smaller *solved* dimension, not an escape from the span; and all x -dependence factors through G , which makes node sampling and exact preassembly available from one decoder.

3.2 PROJECTING THE PDE ONTO THE MANIFOLD

Each PDE we consider gives a discretised residual $r(u)$ that should vanish at the true solution. We substitute the trial manifold $u(z, y)$ into r and project onto a fixed, latent-independent test space: $P \in \mathbb{R}^{M \times n}$ collects the M lowest tensor-product sine vectors of the square, eigenvectors of the five-point operator, $PA = \Lambda P$. The reduced problem solved online is

$$(z^*, y^*) = \arg \min_{z, y} \frac{1}{2} \|r_w(z, y)\|_2^2, \quad r_w(z, y) = \Lambda_\star^{-1} P r(u(z, y)) \in \mathbb{R}^M, \quad M > k + q, \quad (3)$$

with a diagonal row scaling $\Lambda_\star$ stated per PDE in Appendix B: a least-squares Petrov–Galerkin condition with an explicit test space (Carlberg et al., 2011; Lee and Carlberg, 2020), not tangent Galerkin, overdetermined rather than square, solved by a method appropriate to each PDE’s structure.

Elliptic (Poisson). The Poisson residual is $r(u) = Au - f$, and projection gives $r_w(z, y) = B_0(h_\theta(z) + C_q y) - b_0$, $B_0 = PG$ (Appendix B.1); the projector onto $\text{range}(B_0 C_q)$ does not depend on z , so y is eliminated exactly (Golub and Pereyra, 1973), the iteration stays k -dimensional at every q , and $q = R$ is a linear solve.

Parabolic (heat). The heat semi-discretisation $du/dt = -\kappa Au$ is advanced with Crank–Nicolson; the reduced step substitutes the manifold into the fully discrete equation *before* projecting and solves $z_{n+1} = \arg \min_z \|B_0 h_\theta(z) - D B_0 h_\theta(z_n)\|_2$ (Appendix B.2), a nonlinear least-squares problem, not a linear system; at $q = R$ the trajectory is the exact modal propagation of the bank coefficients.

Hyperbolic (Burgers). For $u_t + u(u_x + u_y) = \nu \Delta u$ with a sign-upwind stencil and backward Euler (Appendix B.3) the weak residual is quadratic in the coefficients, so y is not eliminated in closed form; we damp the (z, y) blocks separately inside one Levenberg–Marquardt step (*block-damped* variable projection), which exits stationary everywhere where a joint LM leaves 6 budget exits (Table F.4).

Solver and exits. Each attempt solves the damped normal system $(H + \lambda \text{diag}(\text{diag } H)) \delta = -g$, $H = J^\top J$, $g = J^\top r_w$, accepting the step only if it strictly decreases the residual: damped Levenberg–Marquardt with a monotone test (Marquardt, 1963). Three exit families are recorded and never conflated: the scale-free stationarity measure $\eta(z, y) = \|J^\top r_w\|_2 / (\|J\|_F \|r_w\|_2) \leq \eta_{\text{tol}}$ (4), a residual threshold, and the stalls, reported as early-stopped, never as converged. No query uses the solution it predicts: the elliptic solve starts from the cached training code nearest the projected source, the Burgers query fits (z, y) to the supplied initial field (Appendix B.5).

3.3 HYPER-REDUCTION

The non-linear manifold reduces the DOF count from n to $k + q$, but a projected term such as $PN(Gc)$ still costs $O(n)$. For a fixed linear operator the tested residual is exactly precomputable, $P(Au - f) = B(h_\theta(z) + C_q y) - b$ with $B = \Lambda PG$ assembled offline by two sine transforms, so linear terms are never approximated; the Burgers advection $PN(Gc)$ is the only term that resists preassembly, evaluated *densely*, $O(nR)$ per residual, or on an *empirical quadrature* rule of m nodes (Hernández et al., 2017; Yano and Patera, 2019), $O(mR)$. The rule is a non-negative *per-node* weight vector, *independent of the snapshot*, fitted by NNLS (Lawson and Hanson, 1974) to reproduce the projected advection term at n_{fit} stored codes with a hard cap of m nodes (Appendix B.4); changing m re-solves the fit, rules are not nested, and deployment selects among stored rules. *A rule is never accepted on its NNLS fit residual*; we score it by the held-out relative error ρ of the projected advection term (10) over states the solver actually reaches on trajectories disjoint from the fit and evaluation cases, against a primary bar $\rho_{\text{max}} \leq 0.116$ fixed before any certification job ran and a tight bar 0.06, and then re-draw the construction: a rule is *confirmed* only if every re-draw passes (§6.3).

3.4 MODEL ARCHITECTURE

Three properties of the problem motivate our architecture: the latent iteration is initialised cold, so the decoder Jacobian must retain a well-conditioned linear component, which motivates a *linear skip* in the head; hyper-reduction retains a sparse subset of mesh nodes, so the decoder must evaluate at any node in mesh-independent time, which motivates a *per-node bank*; and the family must reach from

the head’s image to the bank’s whole span without retraining, which motivates the nested *correction directions*.

Linear skip, for cold-start convergence. The head is a two-layer SiLU MLP plus a linear skip, $h_\theta(z) = \varphi_\theta(z) + W^\top z$, so its Jacobian retains a latent-independent component along which the cold solve can descend; the elliptic solver checks the numerical rank of the projected Jacobian at every query. The skip is a design choice, not ablated here.

No encoder; per-node bank. No encoder is needed: the query fits (z, y) against the supplied input. The decoder restricted to a rule’s support is a cached block times the coefficients, and the network’s parameters do not depend on the mesh.

What is fixed, what is chosen, and how error is reported. Every operating point uses one frozen artefact per PDE; selected at run time are the rank q , the quadrature (dense or a stored rule) and the stopping tolerance and budget; M is held fixed along the headline ladder. Every panel reports three numbers: the *bank floor* (projection error onto range G), the *best-found* error (the smallest any point of the augmented manifold attains, from a multistart oracle) and the *solved* error the iteration returns.

4 IMPLEMENTATION

4.1 MATRIX-FREE PROJECTED OPERATORS VIA JAX

Every linear term is preassembled once per mesh as the $M \times R$ matrix B , so the online residual and Jacobian of a linear PDE are $B(h_\theta(z) + C_q y) - b$ and $B[Dh_\theta \ C_q]$, with Dh_θ from a forward-mode Jacobian-Vector Product (Bradbury et al., 2018); the sampled Burgers advection uses the cached stencil block. The damped normal system is solved directly; no Krylov solve is applied to the projected operator.

4.2 TRAINING PROTOCOL

The bank and the head are trained in two stages, both as auto-decoders: the coordinate network g_ϕ is fitted to the training states through (1), then frozen, and the head is fitted with a code library Z on the bank-projected states; C_q is then the principal components of the head’s residual on the training codes. Sizes, cohorts and offline cost are in Appendix C; every headline uses one checkpoint per PDE, named by hash in Table E.1.

5 EXPERIMENTAL SETUP AND BENCHMARK PROBLEMS

All full-order comparators and reduced solves are benchmarked in one Slurm allocation on one GPU per job to provide a direct hardware-to-hardware comparison; *no ratio is ever formed across jobs or GPUs*. Table F.1 specifies each cell and Table F.2 the sampled families.

Problems. *Burgers (hyperbolic, the hero):* $u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$ at 256^2 intervals with a frozen-checkpoint ladder from 64^2 to 1024^2 ; errors are against the same job’s converged full-order solve on the same grid (discretisation error 4.0265 % at 256^2 against a 4096^2 reference). *Poisson (elliptic):* $-\Delta u = f$, ground truth the verified discrete solution from the direct sine transform, at 256^2 and 1024^2 on the square and on the L-shaped domain $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$ at 64^2 to 512^2 , where a sparse direct solve replaces the transform. *Heat and waves:* $\partial_t u - \kappa \Delta u = 0$ with Crank–Nicolson and $u_{tt} = c^2 \Delta u$ with reflective walls.

Metrics and timing. We report the relative L^2 error against the reference of each cell, maximised over cases and output times; for Burgers two maxima are reported everywhere and neither is chosen, worst over *all* times (bounded below by the decoder’s compression of the input at $t = 0$) and worst over the *evolved* times $t > 0$. Every cost is the median over repetitions of a completed computation after a burn-in, arms interleaved in a recorded random order, in float64; “device ms” is the device time with inputs and outputs resident, “complete-query ms” the host-to-host time charging the dense input and output, and every column says which; offline setup is charged separately. Only *admissible*

solves — stationarity 10^{-6} at every step or a residual-rule exit, with a passing quadrature rule — enter a frontier or a ratio, so the 10^{-3} arms are reported but never counted (Table F.13).

Baselines. FNO, a PDEBench-style U-Net and a Transolver are trained on the identical dataset and split at an equal wall budget; only the FNO is timed in the same allocation as the ROM. POD-LSPG, the same projection on a POD basis instead of the trained manifold, runs through the same weak objective, tests, solver and stopping rule. Full-order comparators are always in the same job: preconditioned Newton (Burgers); the direct sine transform and tuned CG (Poisson, heat, waves); sparse direct, CG and IC(0)-PCG (L-shape). The NM-ROM of Kim et al. (2022) is not run under this protocol and is not compared; the earlier comparison against plain CG is kept in Appendix G.6 for continuity. Every table names its job id, GPU and checkpoint (Table E.1), every solve carries its exit reason, and every number is emitted by one generator from audited records.

6 NUMERICAL EXPERIMENTS AND RESULTS

All results here have full tables in the appendix; Table A.1 maps experiments to question, jobs and tables.

Every number is one checkpoint per PDE on development cases unless seeds are stated, and costs compare only inside one allocation. The losses come first.

6.1 COMPARISON AGAINST NEURAL OPERATORS AND FULL-ORDER SOLVERS

Two plain sentences first. At 256^2 and 512^2 a well-tuned full-order solver is cheaper and more accurate than every reduced model, ours included. At 1024^2 3 of the 16 admissible reduced subjects, the cheapest quadrature rungs, are on the non-dominated set, on the evolved-times metric only; the flip lies between 512^2 and 1024^2 . The same-job costs follow (Table 1): on one A100 model the cheapest reduced query costs $4.48\times$ the cheapest full-order setting at 256^2 and $2.92\times$ at 512^2 , a trend on shared hardware; at 1024^2 on an H200 it costs $2.27\times$, a separate fact.

Table 1: The same-allocation panels on Burgers, one job per mesh (Tables F.10, F.12, F.11); same-job ratios only; 256^2 and 512^2 share the GPU model, 1024^2 does not. “Admissible” is the \$5 rule; the 10^{-3} arms are timed in Table F.13 but counted nowhere here.

mesh	job	GPU	timed subjects	admissible reduced (\$5)	non-dom. (evolved)	non-dom. (all-times)	cheapest reduced / cheapest FOM	cheapest reduced / converged FFT	ladder vs-ref span
256^2	3789570	NVIDIA A100 80GB PCIe	48	27	0	0	$4.48\times$	$0.446\times$	$1.13\times$
512^2	3805065	NVIDIA A100 80GB PCIe	48	23	0	0	$2.92\times$	$0.257\times$	$1.36\times$
1024^2	3789572	NVIDIA H200	30	16	3	0	$2.27\times$	$0.191\times$	$1.38\times$

Nothing reduced is on the frontier at 256^2 or 512^2 . Table 1 and Figure 1 come from one allocation on one A100 (job 3789570) holding correction rungs, POD-LSPG, the unrestricted bank, the FNO and full-order Newton. **0 of the 27 admissible reduced subjects are non-dominated** on either metric once the full-order controls are in: the most accurate, POD-512 at 0.2184 % and 2790.8 ms, loses on both axes to 4 same-job full-order settings (cheapest 0.0489 %, 31.8 ms) and the FNO reaches 7.4164 % at 7.2 ms. At 512^2 on the same GPU (job 3805065) 0 of 23 are non-dominated, the frontier is Newton alone and POD-512 (0.3328 %) is beaten on both axes by 4 same-job settings.

At 1024^2 the statement is mesh-qualified. On one H200 (job 3789572) the 3 reduced subjects non-dominated on (GPU ms, worst *evolved* error) are the $q = 0, 16$ and 32 rungs with rules transferred from 256^2 . On *all-times* none is at either mesh, because the $t = 0$ compression (3.71–3.86 % on those rungs) bounds it: which metric is the headline decides whether this is a gain. The transferred rules at $q \geq 64$ miss the primary bar there; why is open.

Neural operators on the same data. Neural operators trained on the same data are more accurate than the reduced model, and the one that was timed against it is also cheaper (Table F.9). Worst error on the matched 8-case cohort: U-Net-small 1.4712, Transolver-refine 1.5224, **ROM** 1.8671, FNO-large 2.4829, efficient full-order solver 0.9978 % (4 operator arms beat the ROM; the FNO’s 7.4164 % above is on the six-case cohort). An earlier FNO-only finding to the contrary is withdrawn; it survives a float64 and a second-seed control (Appendix G.1), all but one operator number is a lower

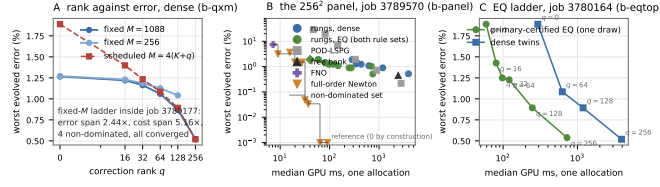


Figure 1: The family on Burgers 256^2 . **A:** worst evolved error against q for the fixed- M and scheduled ladders (b-qxm; no cost axis, costs span jobs). **B:** the same-allocation panel (job 3789570), every admissible subject on (GPU ms, worst evolved error), step line the non-dominated set. **C:** the EQ ladder against its dense twins in one job (job 3780164; rules confirmed at $q = 0, 16, 32$, marginal at $q = 64(2/6), 128(4/5), 256(1/5)$, i.e. passing on some re-draws only).

bound, and the Poisson gap is larger (Table F.33). Their own knob, evaluation resolution, is usable for fno-large only (Table F.31); it moves cost, q moves what the model can represent.

6.2 BEST CONFIGURATIONS PER PROBLEM AND RESOLUTION

The same trained decoder serves every resolution. With one frozen checkpoint per PDE transferred across 64–1024 intervals, the cached reduced solve costs $44.16 \rightarrow 43.58$ ms on Burgers, while the unknowns grow $264\times$ (Table F.26). The complete host-to-host query grows $1.52\times$ through dense input and output. There is no crossover against the cheapest same-job full-order arm at any rung on the square (FOM/ROM 0.265–0.495 on Burgers, 0.063–0.112 on Poisson; Table F.34).

6.3 WHICH KNOB TO TURN

The rank q is the only deployment-time knob moving accuracy. The quadrature rule and the tolerance move cost. Training data and head capacity move the floor but need retraining (Tables F.5, F.19, F.35).

Table 2: The correction ladder on Burgers 256^2 : fixed- M inside job G2 = 3780177 (top) and scheduled $M = 4(K + q)$ in the panel job 3789570 (bottom), development-cohort errors; the scheduled rows add the same checkpoint’s sealed-cohort error (b-seeds job 3804465, the headline), the error against the 4096² reference (“vs ref”; discretisation error 4.03 %) and the EQ rule’s status; costs are per job.

q	M	dev evolved %	sealed evolved %	all %	vs ref %	device ms	EQ evolved %	EQ ms	EQ rule (status)
0	1088	1.2657	—	—	—	848.0	—	—	dense, job 3780177
64	1088	1.0593	—	—	—	1359.4	—	—	dense, job 3780177
128	1088	0.8711	—	—	—	1886.5	—	—	dense, job 3780177
256	1088	0.5194	—	—	—	4377.9	—	—	dense, job 3780177
0	64	1.8890	10.1120	2.5629	4.56	283.9	1.8891	58.6	confirmed (3 of 3 re-draws)
16	128	1.3985	1.4617	2.4806	4.11	360.8	1.4270	81.1	confirmed (3 of 3 re-draws)
32	192	1.2336	1.4458	2.3534	4.08	434.7	1.2493	97.8	confirmed (2 of 2 re-draws)
64	320	1.0843	1.3252	2.1489	4.10	621.1	1.0840	148.9	confirmed (2 of 2 re-draws)
128	576	0.8930	1.0964	1.8116	4.08	1186.9	0.8931	273.5	single-draw
256	1088	0.5194	0.6789	0.9053	4.04	3939.8	0.5129	746.0	single-draw

Rank against test count: the family at fixed M . The rank alone moves the error: hold the test count fixed, add correction directions, and the error falls at every step while the cost rises (Table 2). The ROM approximates the discrete system, so the same-grid error is what the knob controls. Against the fine reference the knob moves the error only from 4.56 to 4.04 % ($1.13\times$), because at 256^2 every rung’s reference error is $1.00\text{--}1.13\times$ the mesh’s own discretisation error of 4.03 %: the reduction adds little on top of the discretisation. And a same-job full-order setting at 15.7 ms reaches 2.47 % against the reference, more accurate than every reduced subject at a small fraction of the top rung’s cost; at 512^2 the ladder runs 3.76 to 2.76 % against the reference over a converged same-grid error of 2.70 %, all 39 reduced subjects above it and the 2 full-order settings below it coarser solves cancelling against the reference, not better ones; at 1024^2 the rungs run 3.86 to 2.80 % over 2.14 %. The knob moves the reduction error, not the physical error. At fixed $M = 256$ the ladder spans only $1.22\times$ and **fails** the pre-registered bar (Table F.5: monotone on evolved, every rung converged, error span $\geq 2\times$, and two generator clauses, ≥ 3 non-dominated points and cost span $\geq 2\times$). At $M = 1088$

the ladder $q = 0, 64, 128, 256$, **inside one job is monotone, every rung converged, spans** $2.44 \times$ **in evolved error for** $5.16 \times$ **in cost with 4 non-dominated points** and passes. $M = 1088$ is the pre-registered pure-rank ladder (the largest fixed M holding every rung to $q = 256$); the $q = 512$ extension did not converge and enters no span; M at $q = 256$ saturates at $M^* = 2176$.

Sealed cohort and seeds (Tables F.30, F.29). The sealed cohort was opened once, after every choice was frozen; on it all 4 checkpoints (the incumbent and 3 fresh seeds) give a ladder monotone on both metrics, 3 of them meeting the knob bar, with top-rung errors of 0.59–0.68 %. The one checkpoint-specific number is the incumbent’s uncorrected $q = 0$ rung, 10.1120 % sealed against 1.8890 % in development (the seeds: 2.15–3.43 %): one sealed case converged to a wrong branch, and the first correction rung removes it ($q = 16$: 1.46–1.87 % on every checkpoint). The pre-registered sealed-over-development ratio thus fails at $q = 0$ for the incumbent alone (5.35; worst seed-mean ratio 1.40); the sealed numbers are the headline and the development table records seed variability (the incumbent beats every seed there at $q = 16, 32, 64, 128$). Rungs not converged everywhere (`seed2` at $q = 64$ (3 budget exits) sealed, the one bar failure; one seed at $q = 256$ in development) are reported as run, not tuned; the fixed- M ladder, one development-cohort checkpoint, is unchanged.

Quadrature and tolerance move cost. Quadrature is a cost lever at equal error ($4.77 \times$ at $q = 0$, $4.81 \times$ at $q = 128$) and the tolerance $10^{-6} \rightarrow 10^{-3}$ removes a further 27–23 % (Table F.10; those arms are not admissible). The earlier ladder’s rule set breaks upward at $q = 256$; the replication-selected set is monotone at both tolerances, $4.2\text{--}5.3 \times$ cheaper than its dense twins, its $q = 128, 256$ rules single-draw (Table 2). Transferred to 512^2 (Table F.12) it stays monotone; its $q = 128$ and 256 transfers pass the primary bar there though single-draw at source, the ladder-rule transfers at $q \geq 64$ do not, and at $q = 256$ the transfer reaches 0.5510 % in $21 \times$ less time than its twin. Solver knobs move cost only (Table F.19).

Validate on reachable states, then re-draw. A quadrature rule must be judged on the states the solver reaches, not on its fit to the states it was built from. The NNLS fit residual says nothing about held-out error (Figure F.1: a rule fitting to $1.5e - 04$ reaches $\rho_{\max} = 0.462$ held-out, $4.0 \times$ the primary bar). **The pre-registered replication re-drew every construction** (Table F.25): $\rho_{\max}$ moves $1.4\text{--}9.6 \times$ within one construction; the ladder’s rules are **confirmed at** $q = 0, 16, 32$ and **marginal at** $q = 64(2/6), 128(4/5), 256(1/5)$; one passing rule does not certify its construction; a re-draw does (Appendix G.2). One gate failed: same-source rule pairs transferred to 512^2 are not bitwise identical (each transfer refits its own draw), and one $q = 32$ source passes the primary bar in one transfer and only the secondary in the other, an $8.7 \times$ spread in $\rho_{\max}$, unexplained; no falsification clause fired.

6.4 THE HEAD AGAINST LINEAR MAPS AT MATCHED DIMENSION

At the same latent dimension the neural head is more accurate than any linear map in the same bank. Per millisecond it is not. At matched $k = 16$ it is compared with the optimal rank- k affine map, a quadratic map, the free bank and POD-LSPG in the same solver (Tables F.17, F.18). On Burgers the head reaches 2.5629 % against 56.9296 % (best linear map) and 61.6503 % (POD-16); on Poisson at 1024^2 it wins per dimension (6.0927 % against 17.2966 %) and loses per ms to POD-128 (4.3452 % at 5.930 vs 5.843 ms). In three layers (Table F.6) the reduction layer sits $6.5 \times$ above the floor, the solver layer costs 0.018 pp on the all-times metric, where the $t = 0$ term dominates: the head binds.

6.5 WHERE THE FAMILY COLLAPSES: LINEAR PDES, NAVIER–STOKES, AND THE L-SHAPED DOMAIN

On linear PDEs there is no trade. The corrections are solved exactly and the top rung is the most accurate and cheapest. On Poisson at 1024^2 (Table F.7) the rungs run $3.1495 \rightarrow 0.9648 \%$ at flat cost, and $q = R$ lands on the floor at 0.7421 % as the *cheapest* point (4.42 vs 3.248 ms exact). Heat and the reflective wave do likewise (Appendix G.5). Against unpreconditioned CG at 10^{-6} , the previous comparator, reduced rungs are $3.7\text{--}23.4 \times$ faster on Poisson in the same job; the transform is faster still by $1.3\text{--}4.0 \times$, hence our comparator (Appendix G.6). **The collapse is confounded with a weak bank:** $3.4\text{--}4.0 \times$ worse than a POD basis of equal rank on both cells. No reduced arm beats the direct solve.

Table 3: L-shaped Poisson, solve layer at $M = 257$, one job per mesh (complete-query ms; ratios inside each job): the cheapest full-order arm, POD-128 and the head at $q = 64$ (full sets: Table F.37).

mesh	job	sparse direct ms	cheapest FOM	ms	err %	POD-128 err %	ms	$\times$ vs direct / cheapest	head $q=64$ err %	ms	$\times$ vs direct / cheapest	head on set
64^2	3784662	1.28	sparse direct (SuperLU)	1.28	0.000	2.591	2.711	0.47 / 0.47	2.300	2.846	0.45 / 0.45	no
128^2	3784662	2.48	sparse direct (SuperLU)	2.48	0.000	2.483	2.764	0.90 / 0.90	2.164	2.806	0.88 / 0.88	no
256^2	3784663	8.39	sparse direct (SuperLU)	8.39	0.000	2.457	2.850	2.94 / 2.94	2.131	3.028	2.77 / 2.77	yes
512^2	3789568	36.50	CG 10^{-2}	29.13	0.385	2.451	4.031	9.06 / 7.23	2.123	4.801	7.60 / 6.07	yes

Navier–Stokes collapses differently. On 2D incompressible flow (nonlinear residual) the family was gated before any ladder (Table F.36): at $K = 16$ and 32 the held-out oracle beats POD- K by only $1.19\times$ and $1.15\times$ against a bar of 2.0 (oracle values are upper bounds), and at $t = 0$ POD-32 is more accurate (POD/oracle 0.82): linear wins where data lie in a low-dimensional subspace. Retraining the head on 128–512 trajectories moves the oracle 27.4938 to 20.6889 to 19.8279 % (gains 25 % then 4 %; “ambiguous” by the pre-registered rule, Table G.1; the bank saw all 512). A family of dimension 8 instead of 14 lowers every error $3.3\text{--}4.4\times$ yet leaves the head’s edge over POD-16 at $1.39\times$ and the gap near 3.8: the edge does not grow as the manifold eases. With $4\times$ the data (2048 trajectories) the gap closes from 4.0 to 1.3 as the head no longer fits the training data: the limit moved to capacity (ratio 1.46). Across $K = 16$ and 32, family dimension 14 and 8, and 128–2048 trajectories, this head class never beats POD- K by the pre-registered $2.0\times$ on held-out decaying 2D Navier–Stokes (ratios 1.15–1.46). An exploratory ladder on that manifold (Table G.2; not phase 3) closes the manifold gap only at $q = R$, where the head no longer matters; POD-LSPG at matched dimension is more accurate at every rung and cheaper below the top, no neural rung non-dominated: the falsification clause’s named outcome. A trade between rungs needs a residual nonlinear in the coefficients and a manifold that beats POD held-out: Poisson, heat and waves lack the first, this cell the second, Burgers had both and still lost on cost. At ten-fold lower viscosity (Appendix G.7) linear floors collapse $12\text{--}19\times$ while the head degrades $3.3\text{--}4.5\times$, every neural rung beats every POD-LSPG rank, $q \geq 64$ is on the reduced-only frontier: the manifold’s edge appears where the Kolmogorov width is worst. Nothing reduced is on the full-order frontier, the fixed- M ladder spans $1.36\times$ and the mesh is under-resolved (discretisation error 19.3–20.9 %): reduced-versus-reduced only.

On the L-shaped domain (Poisson, a linear residual; no fast transform) reduced models are cheaper, POD most of all. The direct solve’s cost climbs with the mesh, at 512^2 CG at tolerance 10^{-2} is the cheaper full-order arm (29.13 ms at 0.385 %, more accurate than any reduced arm), and the reduced arms stay nearly flat (Table 3). Against the cheapest same-job full-order arm the head at $q = 64$ is $2.77\times$ and $6.07\times$ cheaper, POD-128 $2.94\times$ and $7.23\times$; the head is non-dominated only on accuracy (6–19 % more cost than POD-128, 15 % less error). Nothing is claimed beyond 512^2 ; the banks’ reconstruction degrades from 6.8 to 16.7 % on unswept validation sources (Table F.39).

Limitations. (i) We use one checkpoint per PDE. Burgers’ checkpoint is a favourable draw in development (Table F.29), and the fixed- M ladder uses only that seed. A cold-start solve at $q = 0$ can converge to a wrong branch; one sealed case did (10.1120 %, Table F.30). Raising $q \geq 16$ removed it, with no guarantee. (ii) Uniform Cartesian differences, 2D only. Our cohorts are small (Table F.1). The low-viscosity cell is under-resolved. (iii) We ran no cold-start comparison with Kim et al. (2022) and no skip ablation. (iv) Operator numbers are lower bounds (11 of 12 arms still improving). Quadrature rules above $q = 32$ are marginal; one transfer-draw spread in $\rho_{\max}$ is $8.7\times$, unexplained. One cell shows a cross-job spread of 14 %. The 1024^2 frontier rests on one job. (v) Against the fine reference the knob moves the error only $1.13\times$. (vi) No convergence or quadrature-error theory.

7 CONCLUSION AND FUTURE WORK

We built an NM-ROM for elliptic, parabolic and hyperbolic PDEs with a deployment-time accuracy/cost family per model. Rank q and three solver knobs span $2.44\times$ in same-grid error for $5.16\times$ in cost on 2D Burgers in one job. It does not beat a tuned full-order solver or a neural operator on the same data at 256^2 . Where no fast transform applies reduced models are cheaper, plain POD most of all. Next: a resolved low-viscosity cell, unstructured meshes, another Navier–Stokes head class.

REPRODUCIBILITY STATEMENT

Every table and every number in the prose is generated by one script (`paper/gen_tables.py`) from the machine-readable outputs of the runs named in Appendix E; the script records the SHA256 of every file it reads. Each run’s job id, GPU, source commit and checkpoint hash are in Table E.1. Solver definitions, stopping rules and exit-reason codes are in §3.2 and Appendix B; timing and cohort protocol in §5. An anonymised repository with the generator, the provenance registry, the lane summaries and the audit JSONs, and the checkpoints by hash, accompanies the submission; it is built from the paper directory as committed, with nothing outside it, and the header comment of this file is stripped.

AI USE STATEMENT

Language-model assistants were used to draft and edit prose, to write the table generator and the figure scripts, and to audit run records; every number in the paper comes from the recorded run outputs through that generator and none was produced by a language model. The authors take full responsibility for the content.

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A EXPERIMENTS AT A GLANCE

Table A.1 lists every experiment in the paper with the question it answers, the PDE and meshes, the number of cluster jobs and the lane, and where its full numbers are; every row’s jobs, GPUs and checkpoints are in Table E.1.

Table A.1: Experiments at a glance: one row per experiment, the question it answers, the PDE and meshes, the number of cluster jobs (distinct job ids in Table E.1) and the lane, and where the full numbers are.

experiment	question it answers	PDE, mesh	jobs (lane)	where the numbers are
Fixed- M correction ladder	With M held, does the rank q alone move the error, and by how much?	Burgers 256^2	5 (b-qxm)	Tables 2, F.5; Fig. 1A
Rank $\times$ test count	Which of q and M carries the error; where does M saturate?	Burgers 256^2	same (b-qxm)	Tables F.5, F.16
Three-seed retraining	Is the ladder a property of one checkpoint?	Burgers 256^2	3 (b-seeds)	Table F.29
Sealed cohort	Does the ladder hold on cases opened once, after every choice was frozen?	Burgers 256^2	1 (b-seeds)	Tables F.30, 2
Same-job panels	Is anything reduced on the frontier against POD-LSPG, an FNO, a Newton grid and the direct solve in one job?	Burgers 1024^2	256^2 –3 (b-panel)	Tables 1, F.13, F.15, F.14; Fig. 1B
Reference-error column	Does the knob move the physical error or only the same-grid error?	Burgers vs 4096 2	same (b-panel)	Tables 2, F.10, F.12, F.11
Mesh ladder	How do the cached solve and the complete query scale with mesh at a frozen checkpoint?	Burgers, 64^2 – 1024^2 Poisson	2 (mesh-ladder)	Table F.26
Speed at parity	What does the fused implementation cost at bit-level agreement?	Burgers	3 (b-speed)	Table F.34
Solver knobs	Which solver knob moves accuracy and which moves cost?	Burgers	1 (tuning)	Table F.19
Quadrature certification, re-draw	Does the NNLS fit residual predict held-out error; do rules survive re-draws?	Burgers 256^2	3 (b-eqtop)	Tables F.21, F.25, F.22; Fig. F.1
Two rule sets, transfers	Do the two rule sets agree in one allocation; do rules transfer to 512^2 ?	Burgers 512^2	256^2 , same (b-panel)	Tables F.24, F.12; Fig. 1C
Neural operators on shared data	FNO ($\times 3$), U-Net, Transolver, one-variable controls, and the operators’ own knob	Burgers 256^2 ; Poisson	8 (no-second)	Tables F.9, F.33, F.32, F.31; App. G.1
Head ablation, matched dimension	Is the head better than the best linear, quadratic or POD map in the same bank?	Burgers 256^2 ; Poisson 1024^2	2 (head-ablation)	Tables F.17, F.18, F.28; App. G.3
Three-layer decomposition	How much of the deployed error is the bank, the head, and the solver?	every cell	same	Table F.6
Linear PDEs	Poisson ladder, heat linear bank, waves: does the family collapse to a linear model?	Poisson, heat, waves 64^2 – 1024^2	3+1+3 (p-linear, heat, w-ladder)	Tables F.7, F.8, F.27; App. G.5
Navier–Stokes	Phase-2 gate in four settings; head-only data scaling; exploratory ladder	2D decaying NS 256^2	7 (ns2d)	Tables F.36, G.1, G.2, G.3
L-shaped Poisson	Where no fast transform applies, is a reduced solve cheaper than the cheapest full-order solve?	L-shape 64^2 – 512^2	5 (lshape)	Tables 3, F.37, F.39, F.38
Low-viscosity Burgers	Does the manifold’s edge over linear reduction grow where the Kolmogorov width is worst? (under-resolved mesh)	Burgers 256^2	3 (b-lowvisc)	Tables G.5, G.6; App. G.7
Training study	Data density, objective, latent size and smoothness penalty of the head	Burgers 256^2	2 (b-head-train)	Table F.35; App. C
SMA-NM-ROM cold start	Not run under this protocol; a stated limitation	—	0	§6.5

B METHOD DETAILS

B.1 ELLIPTIC INSTANCE: POISSON

For $Au = f$ we take $\Lambda_\star = \Lambda$, so the weak residual is

$$r_w(z, y) = \Lambda^{-1}P(Au - f) = B_0(h_\theta(z) + C_q y) - b_0, \quad B_0 = PG, \quad b_0 = \Lambda^{-1}Pf. \quad (5)$$

Because the retained tests are orthonormal and $Au^\star = f$, this residual is exactly the tested error $P(u - u^\star)$, so the minimised objective is the squared discrete L^2 error of the reduced state in the retained modes; this holds only for the modes P retains.

Analytically eliminated corrections. Let $B_0 C_q = QR$ be a thin QR factorisation. Q does not depend on z , so the inner minimisation over y has the closed form $\mathcal{R}y = Q^\top(b_0 - B_0 h_\theta(z))$ and the minimised value is

$$\min_y \|B_0(h_\theta(z) + C_q y) - b_0\| = \|B_\perp h_\theta(z) - b_\perp\|, \quad B_\perp = (I - QQ^\top)B_0, \quad b_\perp = (I - QQ^\top)b_0. \quad (6)$$

The LM iteration runs on z alone with the deflated operator, and y is recovered by one triangular solve. Because the projector is constant the elimination introduces no approximation and $B_\perp Dh_\theta$ is the exact Jacobian of the eliminated residual; the only dropped derivative anywhere in the elliptic solve is the head curvature. At $q = R$ the deflated operator is zero, the latent problem is degenerate by construction, and the model is the linear least-squares solve $\min_y \|B_0 y - b_0\|$; the timed top rung is that QR solve, not an LM iteration on a zero Jacobian. We report the reduced stationarity of the deflated problem, the augmented stationarity of the full problem, the backward error of the triangular recovery and the numerical rank of $B_0 C_q$.

B.2 PARABOLIC INSTANCE: HEAT

The semi-discrete heat equation $\dot{u} = -\kappa Au$ is advanced with Crank–Nicolson, $(I + \frac{\Delta t \kappa}{2} A)u^{n+1} = (I - \frac{\Delta t \kappa}{2} A)u^n$, and the reduced step substitutes the manifold into the fully discrete equation *before* projecting. With $u^n = u(z_n)$, projecting with P and row-scaling by $\Lambda_\star = I + \frac{\Delta t \kappa}{2} \Lambda$ gives

$$r_{w,n}(z) = B_0 h_\theta(z) - D B_0 h_\theta(z_n), \quad D = \text{diag} \left(\frac{1 - \frac{\Delta t \kappa}{2} \lambda_i}{1 + \frac{\Delta t \kappa}{2} \lambda_i} \right), \quad (7)$$

and $z_{n+1} = \arg \min_z \|r_{w,n}(z)\|_2$, warm-started at z_n . Two points are structural. The right-hand side is built from the *decoded current state* at every step: carrying the quantity the previous step already made small freezes the recursion and reproduces a one-step solution to round-off. And z_{n+1} enters through h_θ , so the step is a nonlinear least-squares problem, not a linear solve. At $q = R$ the step is linear in the coefficients and the whole trajectory is the exact modal propagation of the bank coefficients, with no head and no iteration; this is the “top rung” of the heat and wave ladders. The deployed path factors the damped normal matrix by Cholesky; a variant assembles the Gram matrix $S = B_0^\top B_0$ offline and forms $H = Dh_\theta^\top S Dh_\theta$ directly, which is exact and is reported as an algebraic ablation of the same step.

B.3 NONLINEAR ADVECTION: BURGERS

The full-order problem is $u_t + u(u_x + u_y) = \nu \Delta u$ with the non-conservative sign-upwind stencil

$$N(u)_{ij} = u_{ij} (\delta_x u + \delta_y u)_{ij}, \quad (\delta_x u)_{ij} = N \cdot \begin{cases} u_{ij} - u_{i-1,j}, & u_{ij} > 0, \\ u_{i+1,j} - u_{ij}, & u_{ij} \leq 0, \end{cases} \quad (8)$$

δ_y analogous and switching on the same centre value, ghost zeros on all four walls, and backward Euler in time, $r_n(u) = u - u^n + \Delta t (N(u) - \nu \Delta_h u)$. Projecting and row scaling by $\Lambda_\star = I + \Delta t \nu \Lambda$ gives, with $c = h_\theta(z) + C_q y$,

$$r_{w,n} = (I + \Delta t \nu \Lambda)^{-1} [B_0 c - B_0 c_n + \Delta t (\hat{N}(c) + \nu \Lambda B_0 c)], \quad (9)$$

where the diffusion term is exact and the row scaling is the diagonal of the linearised implicit operator in the modal basis, the modal form of the Helmholtz preconditioner the full-order Newton solver uses. Only $\hat{N}(c) \approx PN(Gc)$ requires a choice: dense evaluation at every interior node, the sampled rule $\hat{N}(c) = \sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(Gc)$ with the sign switch retained at the retained nodes, or the precomputed quadratic form $\frac{1}{2} c^\top Q c$ with Q the symmetrised tensor $T_{iab} = \sum_x P_{ix} G_{xa} (DG)_{xb}$, which is an identity only where $u > 0$ at every stencil node of every trial state the solver visits. The last is an algebraic route we verified in-job (residual and Jacobian parity below 10^{-11} relative on sign-satisfying states) and do not time in this paper. Time stepping uses a fixed substep count per output interval; each step is warm-started at the previous code, and the linear extrapolation is used instead when, and only when, its residual norm is smaller.

B.4 EMPIRICAL QUADRATURE: THE FIT AND THE COUNTS

A fitted rule with support $\mathcal{S}$ and weights w is scored by the held-out relative error of the projected advection term,

$$\rho(u) = \frac{\|\sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(u) - PN(u)\|_2}{\|PN(u)\|_2}, \quad \rho_{\max} \leq 0.116 \text{ (primary bar)}, \quad \rho_{\max} \leq 0.06 \text{ (tight)}, \quad (10)$$

over the held-out reachable states described below; the primary bar is the held-out ρ of the incumbent $q=0$ rule at the state carrying the first-interval penalty, measured in the q-diag cell and adopted as the primary bar in the q-ridge design before any certification job ran; the tight bar was declared before b-eqtop ran. A candidate node set $\mathcal{C}$ is drawn once with a fixed seed. For each of n_{fit} stored codes $c^{(\ell)}$ we evaluate the decoded state, form the per-node contributions $N_i(Gc^{(\ell)})$ and the exact full-grid target $PN(Gc^{(\ell)})$, and solve

$$\min_{w \geq 0} \|\mathcal{G} w - \beta\|_2, \quad \mathcal{G}_{(\ell,i),c} = P_{i,c} N_c(Gc^{(\ell)}), \quad \beta_{(\ell,i)} = [PN(Gc^{(\ell)})]_i, \quad (11)$$

rows normalised by their Euclidean norms, support grown greedily by the Lawson–Hanson criterion with an exact non-negative refit after each addition and a hard cap of m nodes, padded to the requested count if the support saturates. Three counts are separated: the requested node count m , the achieved support, and the number of fit states n_{fit} . In the certification cell the fit states are reachable states (states of the dense solver’s own converged per-step trajectory on fit trajectories), the held-out states are 512 reachable states per rung from certification trajectories disjoint from both the fit and the evaluation cases, and ρ of (10) is evaluated on those. A deployed rule is a stored triple: node indices, weights, and the cached $m \times 5 \times R$ bank block.

B.5 SOLVER CONSTANTS AND EXIT CODES

Damping starts at $\lambda_0 = 10^{-6}$ (Poisson, Burgers) or 10^{-4} (heat); $\lambda \leftarrow \max(\lambda/3, 10^{-12})$ on acceptance and $\min(10\lambda, \lambda_{\text{max}})$ on rejection; the trust radius is taken from the spread of the training codes. Exit codes differ per cell and are reported per cell. Burgers: 4 stationarity, 1 residual, 2 tiny step, 3 rejected at $\lambda \geq 10^{14}$, 0 budget. Poisson: 6 stationarity, 2 relative residual, 1 stall, 3 damping limit, 5 non-finite initial value, 0 budget. Heat: 1 stationarity, 2 tiny step, 3 damping limit, 4 non-finite, 0 budget. The heat criterion divides by $\|J\|_F$ only, applied to a residual already normalised by the target norm; Poisson and Burgers use (4) directly. The completion rule that accepts an attained initial fit (§3.2) was pre-registered in the panel cell’s design before the job ran; the stricter flag is printed beside it in Table F.13.

B.6 WHAT THE METHOD IS, STATED ONCE

For a reader comparing this construction with related descriptions, the following hold for every result here. Dirichlet enforcement is a smooth polynomial factor folded into the bank, with zero lift, and covers homogeneous data only. The projection is least-squares Petrov–Galerkin with a fixed test space, not tangent Galerkin; the two differ even for a linear trial space, since $V^\top A^\top (AVz - b) = 0$ and $V^\top (AVz - b) = 0$ are different conditions. The heat scheme is Crank–Nicolson and the reduced step is a nonlinear least-squares problem, not a linear system in z_{n+1} ; no latent mass matrix is formed and no latent ODE is integrated. The quadrature fit targets the weak advection term with M rows per snapshot; the requested node count m , the achieved support and the fit-state count n_{fit} are three different numbers. No conjugate-gradient solve is applied to the projected operator. There is no encoder in the deployed path. The update is damped Levenberg–Marquardt, not undamped Gauss–Newton. The Poisson ground truth is the verified discrete solution, not an analytic field.

C PER-CELL TRAINING SCHEDULE

This appendix lists the per-cell configuration and training cost referenced in §4. The problem specification per cell (mesh, k , R , M , reference and cohorts) is Table F.1 and the sampled families Table F.2; the offline cost per stage, from the lane records, is Table F.3; the head-retraining arms on the frozen Burgers bank (training-set size, latent dimension and loss terms) are Table F.35. Every headline uses one checkpoint per PDE, named by hash in Table E.1.

D ARCHITECTURE CHOICES: EVIDENCE FROM SWEEPS

This appendix collects the evidence behind the architectural choices in §3.4: the matched-dimension head ablation on Burgers and Poisson (Tables F.17, F.18), the three error layers per cell (Table F.6), the head-capacity sweep on the frozen Poisson bank (Table F.28), the head-retraining arms (Table F.35),

and the solver variants for the corrections on Burgers (Table F.4). The linear skip is a design choice adopted for cold-start convergence and is not ablated in this paper.

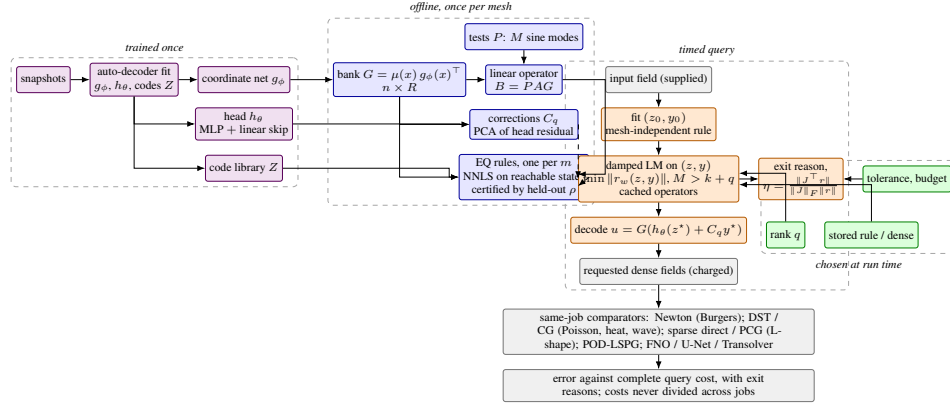


Figure D.1: NM-ROM pipeline. Colour says when each quantity is fixed: purple, trained once and frozen for every result; blue, assembled offline once per mesh; orange, computed inside the timed query; green, chosen at run time among artefacts that already exist; grey, supplied, or a comparator. This is a schematic, not a measurement. The only things that change between operating points are the three green controls (the rank q , the stored quadrature rule or dense evaluation, and the tolerance), so every point of the family comes from one training run.

Figure D.1 draws the pipeline; Table D.1 lists its blocks, their sizes in the symbols of §3.1, and when each is fixed.

Table D.1: The blocks of Figure D.1. “Fixed when” is the colour of the block in the figure.

block	what it is	size	fixed when
coordinate net g_ϕ	random-Fourier-feature MLP, evaluated per node	R outputs per node	trained once
bank G	$\mu(x) g_\phi(x)^\top$ at the mesh nodes, (1)	$n \times R$	assembled once per mesh
head h_θ	two-layer SiLU MLP plus a linear skip	$k \rightarrow R$	trained once
code library Z	auto-decoder codes of the training states	k per state	trained once
tests P	the first M sine modes	$M \times n$	fixed per run
linear operator B	PAG	$M \times R$	assembled once per mesh
corrections C_q	PCA of the head residual on the training codes, nested in q	$R \times q$	once per checkpoint
quadrature rules	NNLS fit on reachable states, one rule per m , validated by held-out ρ	m nodes	once per checkpoint
solve	damped LM on (z, y) (Burgers); y eliminated exactly (linear PDEs)	$k + q$ unknowns	inside the timed query
controls	rank q ; stored rule or dense; tolerance and budget	—	chosen at run time

E PROVENANCE

F FULL TABLES

G EXTENDED RESULTS

G.1 NEURAL OPERATORS: CONTROLS AND THE RESOLUTION KNOB

Both pre-registered controls were run on the selected U-Net (Table F.32): its float64 twin lands at 1.6721 % and its second-seed twin at 1.5479 %, both below the ROM, so that comparison is not a seed or precision artefact — though the float64 margin is 0.1950 pp and, at equal wall clock, the float64 twins ran $0.25\text{--}0.42\times$ of their parents’ epochs, so precision is confounded with an epoch effect. The Transolver arm below the ROM has no twin and remains a single-seed float32 result. Every operator number is a lower bound (7 of 8 of this lane’s arms and 4 of 4 FNO arms were still improving at their budget). The operators’ own knob, evaluation resolution, was tested under a pre-registered rule (Table F.31; speedups are within one model’s own curve in one job and carry no comparison to the ROM or the solver): `fno-large` is usable — one step, $256 \rightarrow 128$, buys the whole gain

Table E.1: Provenance of every result table: job id, GPU, source commit and checkpoint. The SHA256 of every machine-readable file the generator read is in `tables/provenance.json`. Every job asserted `jax_backend=cpu, float64` and highest matmul precision before doing any work.

table	lane	job id(s)	GPU	commit	checkpoint
T3, T5	b-panel (256^2 , bpn301)	3789570	NVIDIA A100 80GB PCIe	f3c5fbed8a21.../7a57f02e2bf4...	incumbent (gate checkpoint_unchanged; hash in T2, tuning row)
T3c, T5c	b-panel (512^2 , bpn401)	3805065	NVIDIA A100 80GB PCIe	f3c5fbed8a21.../7a57f02e2bf4...	incumbent (gate checkpoint_unchanged; hash in T2, tuning row)
T3b, T5b	b-panel (1024^2 , bpn203)	3789572	NVIDIA H200	f3c5fbed8a21.../7a57f02e2bf4...	incumbent (gate checkpoint_unchanged; hash in T2, tuning row)
T4	b-qxm	G1 = 3780175 (NVIDIA A100 80GB PCIe), G2 = 3780177 (NVIDIA A100-PCIE-40GB), S1 = 3780178 (NVIDIA A100-PCIE-40GB), E1 = 3783898 (NVIDIA A100-PCIE-40GB), E2 = 3783899 (NVIDIA A100 80GB PCIe)	per job	ed431edb498a76072b642014	/ 18f0266ae6f04542...
T6a, T7	head-ablation (Burgers)	3711424	NVIDIA A100 80GB PCIe	2718bd320dce...	18f0266ae6f04542...
T6b, T7	head-ablation (Poisson)	3711736	NVIDIA A100 80GB PCIe	6759adcc9d85...	a128e7635c31...
T8	fixed-checkpoint tuning	3712269	NVIDIA A100 80GB PCIe	d2b93bd4f56d...	18f0266ae6f0...
T9	b-eqtop	bet101 = 3780164 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet201 = 3780165 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet301 = 3783811 (NVIDIA A100-PCIE-40GB, commit b2844c607290...)	see job list	see job list	18f0266ae6f04542...
T10	mesh-ladder (Burgers)	3711388	NVIDIA A100-PCIE-40GB	521cdced6f4d...	18f0266ae6f0...
T10	mesh-ladder (Poisson)	3711389	NVIDIA A100 80GB PCIe	521cdced6f4d...	a128e7635c31...
T21	p-linear + lshape (secondary: previous comparator, CG)	the p-linear and lshape jobs above	per job	per job	same checkpoints as T11b / T18c
T20, T20b	b-lowvisc (appendix; F4 under-resolution caveat)	panel 3817807; gate 3789639; training 3804337	NVIDIA A100-PCIE-40GB	2cf5dc34	low-viscosity checkpoint hashed in the lane summary
T11e	ns2d phase 2 (K=16, K=32, family dimension 8, 2048 trajectories); lane closed, no phase-3 job	3787319 (K=16, R=256), 3787320 (K=32, R=512), 3808498 (K=16, R=256, family dimension 8), 3808495 (K=16, R=256, 2048 trajectories); FOM 3780151	ns303 A100 (pax105); others per job	per job	checkpoints hashed in each result.json
T11f	ns2d ns301 (head-only data scaling on the frozen K=16 bank)	3808493	per job	per job	frozen K=16 bank; heads hashed in result.json
T11g, T11h	ns3d ns304 (exploratory after a failed phase-2 gate)	3808502	NVIDIA A100 80GB PCIe	31e0846f	ckpt_K32_R512 hashed in result.json
T11a	w-ladder	64 ² : job 3780447 (NVIDIA A100 80GB PCIe, commit 0bb3cc86); 256 ² : job 3783805 (NVIDIA A100-PCIE-40GB, commit 2655bb01); 1024 ² : job 3780450 (NVIDIA A100 80GB PCIe, commit 0bb3cc86)	per job	per job	frozen-math SHA asserted in job
T11d	heat linear bank (2026-09-10)	3511417	NVIDIA A100-PCIE-40GB	73fdaa88eb75...	expanded_seed790715 (frozen)
T11b, T11c	p-linear	256 ² : job 3780692 (NVIDIA A100-PCIE-40GB, commit b43a437d7360); 1024 ² : job 3783813 (NVIDIA H200, commit 3e411b5ac59d); head capacity job 3783883	per job	per job	R=512/K=32 checkpoint (pbb02 primary)
T14, T14c, T14d	no-second (5 of 8 jobs counted; two preamble deaths uncounted)	3702464, 3702709, 3710846, 3780138, 3780139, 3780625, 3783831, 3787189	A100 (per job)	per job	operator checkpoints hash-verified in job
T15	b-speed	3745655 (spd01), 3745656 (fine01), 3745913 (comp01)	A100 80GB PCIe	8fdffb08 / 94399dd6	18f0266ae6f04542...
T16	b-head-train	3745912 (training), 3749074 (evaluation)	A100-PCIE-40GB	0f0c56f7 / 2b9e7ee7	trained checkpoints hashed in archive
T18, T18c, T18d	lshape	training 3783786; solves 3784662, 3784663, 3789568; free rung 3784910	NVIDIA A100 80GB PCIe	1086ccefcd5...	7 heads + bases Git-tracked
T12	b-seeds (development cohort)	seed1 = 3783776 (NVIDIA A100 80GB PCIe); seed2 = 3783777 (NVIDIA A100 80GB PCIe); seed3 = 3783778 (NVIDIA A100-PCIE-40GB)	per job	per job	three seed checkpoints hashed in summary
T13	b-seeds (sealed cohort)	3804465	NVIDIA A100-PCIE-40GB	be9415ab	four checkpoints (incumbent + three seeds) hashed in summary

($2.12\times$ its own speed at $1.14\times$ its own error), below which the query is launch-bound (3.46, 3.44, 3.33 ms) while the error rises to $1.54\times$ and $2.17\times$ — while `tsol-refine`, `unet-refine` are not (the U-Net breaks at once, the Transolver is cost-flat). At rung 128 no family is within $32\times$ of the interpolation floor, so the off-resolution error is the network's, not the grid's.

G.2 QUADRATURE: THE FIT-STATE COUNT AND THE REPLICATION

Scored by (10) on reachable states, the top-rung blocker is the number of fit states, not the node count: 14- and 8-state rules at $q = 128$ and 256 fail at $m = 2048$ ($\rho_{\max} = 0.1908, 0.1678$) where 64-state rules pass in that draw (0.0669, 0.1074; Table F.21). Rebuilt with the cheapest rule passing the primary bar per rung and timed in one allocation, the EQ ladder runs $1.8891 \rightarrow 0.5389\%$ at $59.1 \rightarrow 722.2$ ms (Table F.24; offline cost in Table F.3). In the replication (pool and fit-state subset,

Table E.2: Attempts that produced no reported number. A retracted attempt contributes no timed row, no gate and no oracle value; it is listed so that it is visibly retracted rather than absent.

lane	attempt	job	what it would have produced	why nothing is reported
b-panel	bpn201	3783817	1024 ² panel	config-parsing bug; no timed number produced
b-panel	bpn202	3787247	1024 ² panel	out of memory during autotuning inside an untimed diagnostic
lshape	lsh01	3780148	bank and head training	mis-scaled basis-orthonormality gate; rerun in full as lsh02
p-linear	plin1024	3780691	Poisson 1024 ² ladder	GPU out of memory in the untimed best-found oracle; rerun on an H200
w-ladder	wl256b	3780448	wave 256 ² ladder	retained-value gate failed on a tie-breaking difference; rerun as wl256c
no-second	pois01	3780224	Poisson U-Net screen	stager omitted a config directory; no training ran; rerun as pois02
ns2d	ns202	3783797	Navier-Stokes $K = 32$ head	pre-§A4 attempt on a rank-capped bank; superseded by ns204

Table E.3: Attempts in flight when this version was prepared: none; every lane this version reads is closed.

lane	attempt	job	what it will add
—	—	—	none: every lane read by this version is closed

four draws at fixed q , m , fit-state count; Table F.25) the rules are confirmed at $q = 0, 16, 32$ and marginal at $q = 64(2/6)$, $128(4/5)$, $256(1/5)$ (draws passing / draws); at $q = 256$ the rule the ladder ran is the only one of 10f5 draws of its recipe to meet the bar (re-draws $0.130 - -0.242$). “Every rung carries a certified rule” is withdrawn as a statement about the construction, and four earlier readings are withdrawn by the same data.

G.3 THE HEAD AT MATCHED DIMENSION: METRIC DEPENDENCE AND THE THREE LAYERS

On Burgers, on the all-times metric no POD rank up to $k' = 128$ matches the head (10.1198 %), while on evolved times POD-128 reaches 1.9464 % against the $q = 0$ rung’s 1.8890 % (both on the development cohort; the rung’s sealed value is 10.1120 %, Table F.30). In three layers (Table F.6): floor 0.3918 %, best-found 2.5447 %, solved 2.5629 %. On Poisson a wider head moves the best-found/floor ratio only from 4.18 to 2.76 (Table F.28).

G.4 NAVIER-STOKES: THE HEAD-ONLY DATA-SCALING DIAGNOSIS

Job 3808493 retrained only the head on the frozen $K = 16$, $R = 256$ bank with 128, 256 and 512 training trajectories under three recipes and scored the held-out oracle at 256^2 (Table G.1). The pre-registered rule (ns2d/DESIGN.md §A9 (pre-registered data-scaling rule for the head-only diagnosis): least-squares log-log slope of the held-out oracle in the training-trajectory count; steeper than $-0.25 = \text{data-limited}$, shallower than $-0.10 = \text{head-limited}$, inside the band = ambiguous) returns “ambiguous” for every recipe; the decelerating slope and the inertness of regularisation lean towards a head-limited reading, which the lane does not upgrade beyond ambiguous. Caveat, pre-registered: the frozen bank was trained on all 512 trajectories, so the small- n arms are optimistic.

Exploratory ladder after the failed gate (not a phase-3 result). Every number in this paragraph and in Tables G.2 and G.3 is exploratory after a failed phase-2 gate: the $K = 32$, $R = 512$ manifold failed the held-out oracle bar (Table F.36), so the ladder answers a mechanism question only, not the pre-registered phase-3 question. Job 3808502 (NVIDIA A100 80GB PCIe) ran the correction ladder $q \in \{0, 32, 64, 128, 256, 512\}$ at one fixed test count $M = 2176$ on 8 development cases with 3 timed repetitions, same-job POD-LSPG at every matched dimension $k' = K + q$, and a full-order Newton-tolerance ladder in the same job. The correction rank closes the manifold gap: the manifold layer (median over cases of the worst evolved time) falls from 16.4787 to 1.7434 %, the bank floor (1.7434 %), but reaches it only at $q = R = 512$, where the head no longer matters; the residual directions capture 26, 44, 68, 92, 100 % of the head-residual energy at $q = 32, 64, 128, 256, 512$. On the matched statistic (the same median-over-cases of the worst evolved time on both sides) the solve layer is $1.4\text{--}2.5\times$ the manifold layer, decreasing with q from 2.51 at $q = 0$ to 1.40 at $q = R$ (an earlier median over all 48 states, $t = 0$ included, gave $2.9\text{--}10.8\times$ and is superseded: it compared two different statistics). The loss accumulates along the trajectory: at $q = 0$ the per-time solved/manifold ratio runs 1.44, 1.66, 2.19, 2.89, 3.85 from $t = 0.2$ to $t = 1$. The ROM’s enstrophy over the reference

Table F.1: Problem specification. Cohort and reduced sizes are read from the run configurations where recorded; the sealed final cohorts have not been opened for any cell.

PDE	equation, domain, boundary	meshes	time stepping	reduced sizes	reference	cohorts
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 (ladder 64–1024)	$\Delta t = 0.005$, backward Euler, sign- upwind	$K = 16$, $R = 512$	refined 4096 ² , $\Delta t = 0.00015625$	6 development cases; 32 held-out (tuning); sealed cohort opened once (job 3804465)
Poisson 2D	$-\Delta u = f$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	$256^2, 1024^2$	none (elliptic)	$K = 16$, $R = 128$ (incumbent); $K = 32$, $R = 512$	exact discrete (DST); 2048 ² refinement	12 development sources
Heat 2D	$u_t = \kappa \Delta u$, $(0, 1)^2$	64^2 – 1024^2	Crank–Nicolson	$k = 8$, $R = 32$	exact modal	12 development cases (ear- lier cell, job 3511417)
Wave 2D (reflec- tive)	$u_{tt} = c^2 \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	$64^2, 256^2$, 1024^2	RK4 on the manifold; exact modal propaga- tion for the bank	$K = 32$, $R = 64$	direct DST	8 development cases
Poisson, L-shape	$-\Delta u = f$, $(0, 1)^2 \setminus [\frac{1}{2}, 1]^2$	$256^2, 512^2$	none	$K \in \{16, 32\}$, $R \in \{256, 512, 514\}$	sparse direct (Su- perLU)	3072 / 256 / 32 sources (train / selection / develop- ment)

Table F.2: Sampled problem families, transcribed from the generator sources named in the last column (code constants, not run outputs).

PDE	equation and boundary	sampled family (transcribed from the generator source)	source
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$u_0 = a \exp(- x - c ^2/2a^2)$, $c_1 \sim U(0.15, 0.85)$, $w \sim U(0.05, 0.20)$, $a \sim U(0.5, 2)$; $\nu \sim \log U(0.01, 0.1)$; outputs $t \in \{0.05, \dots, 0.25\}$	experiments/mr-burgers2d/engines.py:params_draw
Poisson 2D	$-\Delta u = f$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$f = a \exp(- x - c ^2/2a^2)$, $c_1 \sim U(0.15, 0.85)$, $w \sim \log U(0.02, 0.1)$, $a \sim U(0.5, 2)$	multistage-precision/ms_parametric.py:sample_params
Poisson, L-shape	same source family on $(0, 1)^2 \setminus [\frac{1}{2}, 1]^2$	as above; sources centred in the removed quadrant rejected	experiments/lshape
Heat 2D	$u_t = \kappa \Delta u$ on $(0, 1)^2$, $\kappa = 0.02$ fixed	polynomial-boundary Gaussian family (single_bc_poly_gaussian.v1); Crank–Nicolson $\Delta t = 0.025$	heat-linear-bank-report_job_3511417
Wave 2D (reflec- tive)	$u_{tt} = c^2 \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	compact bump $\times$ Gaussian: half-widths $s_1 \sim U(0.36, 0.42)$, centre $c_1 \sim U(s_1 + 0.025, 1 - s_1 - 0.025)$, amplitude $\sim U(0.7, 1.3)$, $\sigma_s \sim U(0.12, 0.16)$, advective velocity $v_1 \sim U(-0.5, 0.5)$ (seen every fourth case); speed $c \sim U(0.85, 1.15)$	experiments/multiresolution-wave/audit_dynamics.py:parameter_rows

at $t = 1$ is 1.043 (median) at $q = 0$, 0.990 at $q = 32$, a deficit of 0.95–0.97 at $q = 64$ –256 and 0.999 at $q = R$ (POD-32 control 1.026). Zero budget exits: worst evolved 68.7654 $\rightarrow$ 6.0140 % (11.4 $\times$ for 3.9 $\times$ in cost), monotone in the median (yes) but not in the worst (no; one inversion at the first rung). POD-LSPG at the same dimension is more accurate at every rung (yes) and cheaper at 5 of 6 rungs (all but $q = 512$, where both cost about the same); the non-dominated set on (ms, worst evolved) is fom_ntol0.0001, fom_ntol0.001, fom_ntol0.003, fom_ntol0.03, fom_ntol1e-06, fom_ntol1e-11, pod_k32 — 0 neural rungs, the falsification clause’s named outcome for this head. The loosest full-order setting that converges reaches 0.0041 % at 421.3 ms (converged reference 1177.7 ms).

G.5 THE LINEAR CELLS IN FULL

Poisson at 1024^2 ($R = 512$, $K = 32$; Table F.7): the middle rungs cost a flat 6.4–6.9 ms; the pre-registered degenerate-curve criterion passes (at 256^2 only its literal cost-span clause fails, because the top rung is 3.17 $\times$ cheaper than the middle rungs); solver effort buys nothing at $q = 0$ (floor 0.7421, best-found 3.1139, solved 3.1495 % at $M = 129$). Heat at 1024^2 (an earlier cell of the same family, Table F.8): the bank evolved linearly reaches 1.68 % at 0.56 ms against the head’s 4.56 % at 12.3 ms, with the direct solve at 1.03 ms. The reflective wave (Table F.27): the top rung, the bank under exact modal propagation with no head, is not strictly the most accurate rung (strict reading: no) but matches the $q = 32$ rung within the pre-registered integrator tie band of 0.020 pp at every mesh (256^2 : 5.121 against 5.104 %), and is 158–42 $\times$ cheaper than the cheapest head rung on device ms (on complete-query ms the dense input and output dominate both, Table F.27); the head ladder is monotone in q (yes), cost is not. On waves POD $k' = 64$ reaches 1.502 % against the bank’s 5.121 % at the same cost. On the L-shaped domain the sparse direct solve costs 1.28, 2.48, 8.39, 36.50 ms at 64^2 , 128^2 , 256^2 , 512^2 ; at 64^2 it dominates everything and at 128^2 only POD rungs join the non-dominated set (POD $k'=64$ at 7.701 %, 1.08 $\times$ the direct solve’s cost), cheapness rather than a trade; at 512^2 the best POD rung, POD $k'=128$, reaches 2.451 % at 4.031 ms. In a separate job at $M = 1024$, whose costs are not comparable with those above, the free rung $q = R$ is the one place the learned bank beats the linear baseline on both axes, at 1.003 $\times$ its own bank floor with no solver iterations (Table F.38).

G.6 AGAINST THE PREVIOUS SUBMISSION’S COMPARATOR: CONJUGATE GRADIENT

An earlier version of this work compared the reduced solve against unpreconditioned conjugate gradient (CG) at tolerance 10^{-6} . That comparison is retained here as a secondary one, same job only (Table G.4): on Poisson the reduced rungs are 3.7–23.4 $\times$ faster than CG 10^{-6} in the same job, but the direct transform (DST) solve is faster than every rung by 1.3–4.0 $\times$, which is why the paper

Table F.3: Offline cost per stage, from the lane records: head training on the retrained arms (the incumbent’s own training time was not recorded), the correction-direction fit, and the NNLS fit of each rule the primary EQ ladder ran. Every rule is refit when m changes; nothing is nested.

offline stage	seconds	amortised over	source
head training, like-for-like retrain of the incumbent recipe (4608 traj., 200k steps)	387	once per checkpoint	b-head-train job 3745912 (A100-PCIE-40GB)
head training, selected retrained arm	1459	once per checkpoint	b-head-train job 3745912 (A100-PCIE-40GB)
correction directions C_q (PCA of the head residual; flattened LM fit of the best-found codes)	1242	once per checkpoint, all q	cheap-corrections job 3734098
certified EQ rule, $q = 0$, $m = 1024$ (reachable, qrg304:reachable)	72	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 16$, $m = 1024$ (reachable, qrg304:reachable)	147	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 32$, $m = 1024$ (reachable, qrg304:reachable)	155	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 64$, $m = 1024$ (reachable, qrg304:reachable)	160	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 128$, $m = 2048$ (fs64, this_job:reachable)	2920	once per rung and rule	b-eqtop job 3780164
certified EQ rule, $q = 256$, $m = 2048$ (fs64, this_job:reachable)	2650	once per rung and rule	b-eqtop job 3780164

Table F.4: How the corrections are solved on Burgers, one job (job 3734098, NVIDIA A100 80GB PCIe): joint LM on (z, y) , block-damped, and plain variable projection at $q = 64$ and 128. Same error where all converge; the block-damped step is the one kept.

arm	M	worst all-times %	budget exits	all stationary	device ms
$q=64$, joint LM on (z, y)	320	2.1489	6	no	1119.7
$q=64$, block-damped	320	2.1489	0	yes	619.2
$q=64$, plain variable projection	320	2.1489	0	yes	9527.9
$q=128$, plain variable projection	256	1.8116	297	no	47566.9
$q=128$, block-damped	256	1.8116	0	yes	825.5

compares against the transform. On the L-shaped domain, where no transform applies, the lane timed CG at its own tolerances (tightest 10^{-10}) and an IC(0)-preconditioned CG on the CPU: the head is $2.9\text{--}14.9\times$ faster than the tightest CG across 64^2 , 128^2 , 256^2 , 512^2 , and the sparse direct solve is the competitive comparator (Table 3).

G.7 BURGERS AT TEN TIMES LOWER VISCOSITY (UNDER-RESOLVED; REDUCED-VERSUS-REDUCED ONLY)

A second Burgers cell (lane b-lowvisc; training job 3804337, gate job 3789639, panel job 3817807 on NVIDIA A100-PCIE-40GB) keeps the incumbent recipe and mesh and lowers the viscosity range ten-fold. The pre-registered under-resolution clause (F4) applies: the converged discrete operator itself sits $19.3\text{--}20.9\%$ from the 4096^2 reference ($5.17\times$ the incumbent cell’s discretisation error), so every number below is evidence about reduced models against each other on the same discrete system, not about the PDE. Two things did happen. Linear reduction collapses: the POD-512 floor degrades $12.09\times$ and the trained linear bank’s floor $19.00\times$ ($0.3918 \rightarrow 7.4455\%$) against the incumbent cell, while the neural head’s solved $q = 0$ error degrades $3.27\times$ ($2.5629 \rightarrow 8.3933\%$) and its best-found $4.48\times$ (an upper bound here: solved over best-found is 0.74); the pre-registered F2 clause therefore does not fire (no). And every neural rung of the fixed- $M = 1088$ ladder, from $q = 0$, beats every POD-LSPG rank $k' = 16\text{--}512$ (best 11.2540% ; not monotone past $k' = 256$, the $k' = 512$ solve being solver-limited), and from $q = 128$ up also the free bank’s own solve (7.7940% at 3174.6 ms): on the reduced-only frontier the neural rungs $q = 64, 128, 256$ are non-dominated. What did not happen: no reduced subject is non-dominated against the full-order grid (criterion P fails; the non-dominated set is nt1e-2_dt01, nt1e-3_dt005, dense_tight, fft_tight; the cheapest full-order setting beating every reduced subject is nt1e-2_dt01 at 14.0 ms and 3.90%), the fixed- M ladder is monotone and converged but spans only $1.36\times$ in error for $3.25\times$ in cost ($9.0500 \rightarrow 6.6718\%$, $557.4 \rightarrow 1810.3\text{ ms}$), so the $2\times$ knob bar is not met, the $M = 256$ control is not monotone (test-starved at $q = 128$, as in the incumbent cell), and every reduced row’s error against the reference ($17.98\text{--}19.44\%$) is the discretisation error. 27 subjects, 0 budget exits.

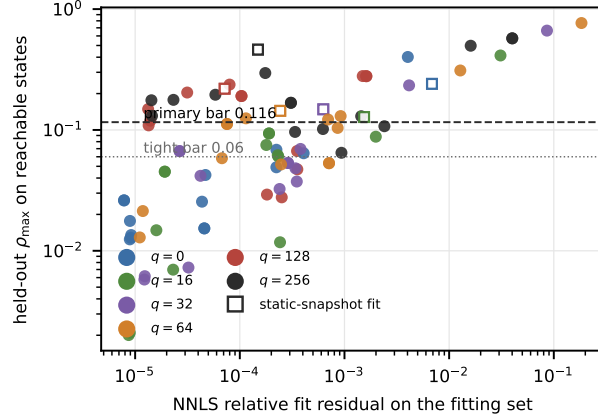


Figure F.1: Quadrature rules: fit residual against held-out error. Each point is one fitted rule; the horizontal axis is its NNLS fit residual on the states it was fitted to, the vertical axis its worst held-out ρ (10) on states the solver actually reaches, with the primary and tight bars drawn. Filled markers are rules fitted on reachable states, hollow squares rules fitted on static snapshots. Data: every rule of the b-eqtop lane, jobs 3780164, 3780165, 3783811 (Table F.22). What it shows: the fit residual does not predict the held-out error, and because the same construction re-drawn moves by $1.4\text{--}9.6\times$ (Table F.25), a point below the bar is a passing draw, not a certified construction.

Table F.5: Rank against test count (dense advection, budget 600, single seed). Both fixed- M ladders are shown; the $M = 256$ ladder fails the bar and the $M = 1088$ ladder passes it, and $M = 1088$ was chosen after the grid was run. The scheduled ladder’s cells span two jobs so its costs are not printed. Jobs G1 = 3780175 (NVIDIA A100 80GB PCIe), G2 = 3780177 (NVIDIA A100-PCIE-40GB), S1 = 3780178 (NVIDIA A100-PCIE-40GB), E1 = 3783898 (NVIDIA A100-PCIE-40GB), E2 = 3783899 (NVIDIA A100 80GB PCIe); commit ed431edb498a / 76072bf42014; checkpoint 18f0266ae6f04542. ...

ladder	q	M	worst evolved %	device ms
fixed $M = 256$ (job 3780177)	0	256	1.2710	406.2
fixed $M = 256$ (job 3780177)	64	256	1.1255	645.6
fixed $M = 256$ (job 3780177)	128	256	1.0418	921.8
fixed $M = 1088$ (job 3780177)	0	1088	1.2657	848.0
fixed $M = 1088$ (job 3780177)	64	1088	1.0593	1359.4
fixed $M = 1088$ (job 3780177)	128	1088	0.8711	1886.5
fixed $M = 1088$ (job 3780177)	256	1088	0.5194	4377.9
scheduled $M = 4(K + q)$	0	64	1.8890	—
scheduled $M = 4(K + q)$	16	128	1.3985	—
scheduled $M = 4(K + q)$	32	192	1.2336	—
scheduled $M = 4(K + q)$	64	320	1.0843	—
scheduled $M = 4(K + q)$	128	576	0.8930	—
scheduled $M = 4(K + q)$	256	1088	0.5194	—

H GLOSSARY

Written for a reader who knows none of this project’s vocabulary.

Bank, head, latent code ($G, h_\theta, z; R, k$)

The fixed spatial functions the decoder combines (width R); the small network that maps k latent numbers to their coefficients; the k numbers solved for per query.

Correction rank q , rung, ladder, C_q

The number of fixed extra bank directions the solver may add to the head’s output; one value of q ; the sequence of rungs; the matrix of directions, a nested prefix chosen offline. $q = 0$ is the head alone, $q = R$ the linear model on the bank.

Table F.6: Three layers of deployed error, worst over each cell’s development cases: what the bank can express, what the augmented manifold can reach (multistart oracle, an upper bound), what the deployed iteration returned. The p-linear $q > 0$ oracle was mis-scaled and is retracted; only its $q = 0$ oracle appears.

cell	arm	bank floor %	best-found %	solved %	best/floor	solved – best (pp)	source
Burgers 256^2	neural head, $q=0$	0.3918	2.5447	2.5629	6.49	0.018	head-ablation
Poisson 1024^2 , $R=128$	neural head	2.3148	6.0926	6.0927	2.63	0.000	head-ablation
Poisson 256^2 , $R=512$	neural head $K=32$, $q=0$	0.7459	3.1211	3.1217	4.18	0.001	p-linear
Poisson 1024^2 , $R=512$	neural head $K=32$, $q=0$	0.7421	3.1139	3.1146	4.20	0.001	p-linear
wave 256^2	head, $q=0$	2.4571	8.0772	11.3314	3.29	3.254	w-ladder
wave 256^2	head, $q=32$	2.4571	2.4571	5.1038	1.00	2.647	w-ladder
L-shape 256^2	head_sdf_R512_K32	0.7766	3.1802	3.1814	4.09	0.001	lshape (untimed)
L-shape 256^2	head_sdf_R512_K16	0.7766	3.8607	3.8608	4.97	0.000	lshape (untimed)

Table F.7: Poisson ladder at 256^2 and 1024^2 (one job per mesh: 256^2 : job 3780692 (NVIDIA A100-PCIE-40GB, commit b43a437d7360); 1024^2 : job 3783813 (NVIDIA H200, commit 3e411b5ac59d); never compare costs across meshes; $q = 512$ via LM is degenerate by construction and its QR twin is the timed top rung).

mesh	subject	M	worst %	median %	complete-query ms	device ms	valid	non-dom. (all)	non-dom. (reduced)
256^2	$q=0$	129	3.1567	0.5080	9.214	6.703	36/36	no	no
256^2	$q=32$	257	2.4699	0.4437	9.537	6.893	36/36	no	no
256^2	$q=64$	384	2.0808	0.4127	9.819	7.347	36/36	no	no
256^2	$q=128$	640	1.5497	0.3195	10.219	7.536	36/36	no	no
256^2	$q=256$	1153	0.9689	0.1315	9.658	7.121	36/36	no	no
256^2	$q=512$	2177	0.7459	0.0580	117.854	114.990	0/36	no	no
256^2	linear top rung ($q=R$, QR)	2177	0.7459	0.0580	3.221	1.693	36/36	no	yes
256^2	free bank (LM)	2177	0.7789	0.0597	23.295	21.416	36/36	no	no
256^2	POD $k'=32$	129	9.9912	2.7371	3.766	2.094	36/36	no	no
256^2	POD $k'=64$	257	5.7375	0.6105	4.062	2.309	36/36	no	no
256^2	POD $k'=128$	512	2.5117	0.1060	4.825	3.231	36/36	no	no
256^2	POD $k'=256$	1025	0.8165	0.0134	6.171	4.566	36/36	no	no
256^2	POD $k'=512$	2048	0.1855	0.0014	10.002	8.340	36/36	no	yes
256^2	direct DST	—	0.0000	0.0000	2.526	0.227	36/36	yes	no
256^2	CG 10^{-2}	—	0.1527	0.0554	23.818	22.290	36/36	no	no
256^2	CG 10^{-4}	—	0.0012	0.0003	32.698	30.962	36/36	no	no
256^2	CG 10^{-6}	—	0.0000	0.0000	38.046	36.330	36/36	no	no
1024^2	$q=0$	129	3.1495	0.5079	6.394	3.941	36/36	no	no
1024^2	$q=32$	257	2.4641	0.4437	6.592	4.108	36/36	no	no
1024^2	$q=64$	384	2.0760	0.4128	6.745	4.240	36/36	no	no
1024^2	$q=128$	640	1.5459	0.3195	6.875	4.393	36/36	no	no
1024^2	$q=256$	1153	0.9648	0.1315	6.849	4.319	36/36	no	no
1024^2	$q=512$	2177	0.7421	0.0580	43.220	40.632	0/36	no	no
1024^2	linear top rung ($q=R$, QR)	2177	0.7421	0.0580	4.424	1.830	36/36	no	yes
1024^2	free bank (LM)	2177	0.7751	0.0596	13.021	10.662	36/36	no	no
1024^2	POD $k'=32$	129	9.9816	2.7357	3.696	1.318	36/36	no	no
1024^2	POD $k'=64$	257	5.7282	0.6099	3.970	1.571	36/36	no	no
1024^2	POD $k'=128$	512	2.5019	0.1058	4.397	2.095	36/36	no	no
1024^2	POD $k'=256$	1025	0.8119	0.0133	5.262	2.908	36/36	no	no
1024^2	POD $k'=512$	2048	0.1838	0.0014	6.972	4.572	36/36	no	yes
1024^2	direct DST	—	0.0000	0.0000	3.248	0.237	36/36	yes	no
1024^2	CG 10^{-2}	—	0.0722	0.0297	62.498	60.017	36/36	no	no
1024^2	CG 10^{-4}	—	0.0004	0.0001	85.988	83.437	36/36	no	no
1024^2	CG 10^{-6}	—	0.0000	0.0000	103.642	101.260	36/36	no	no

Test count M , weak tests, row scaling

How many sine test functions the residual is averaged against; the functions; the diagonal weights on those averaged equations. All three change which solution is selected.

m , EQ, rule, NNLS, fit states, reachable states

Quadrature node count; empirical quadrature, evaluating a sum over all nodes by a weighted sum over m of them; one fitted instance; the non-negative least-squares fit; the states used to fit a rule; states the solver actually visits, as opposed to stored snapshots.

ρ , primary and tight bars, confirmed, marginal

The rule’s relative error on the projected advection term on held-out reachable states; $\rho_{\max} \leq 0.116$ and ≤ 0.06 ; a construction every one of whose re-draws passes; one only some of whose re-draws pass. Never the NNLS fit residual, and never a single draw.

Dense

Evaluating the advection term at every interior node, exactly.

Evolved vs all-times metric, $t=0$ compression

Worst error over output times $t > 0$; over all output times including $t = 0$; the error at $t = 0$ alone, which is the decoder’s reconstruction of the supplied initial field and bounds the all-times metric from below.

Table F.8: Heat 2D, an earlier cell of the same decoder family (job 3511417, NVIDIA A100-PCIE-40GB, commit 73fdaa88eb75...; $k = 8$, $R = 32$, twelve development cases, three repetitions; the finest three meshes of a five-rung ladder). The linear evolution of the full bank is the $q = R$ rung.

mesh	arm	worst physical %	worst same-grid %	device ms	complete-query ms
64 ²	linear bank, exact modal evolution ($q=R$, no head)	1.675754	1.675754	0.116167	1.146627
64 ²	nonlinear head ($k=8$)	4.559260	4.554246	12.091245	12.540239
64 ²	same-grid direct solve	0.089749	0.000000	0.184825	0.557447
64 ²	coarse (16 ²) direct solve, interpolated	2.585355	2.623291	0.176126	0.518843
256 ²	linear bank, exact modal evolution ($q=R$, no head)	1.675830	1.675830	0.137731	2.326213
256 ²	nonlinear head ($k=8$)	4.555681	4.555390	11.888083	13.147866
256 ²	same-grid direct solve	0.005602	0.000000	0.209627	1.438594
256 ²	coarse (16 ²) direct solve, interpolated	2.578601	2.581196	0.190585	1.400465
1024 ²	linear bank, exact modal evolution ($q=R$, no head)	1.675830	1.675830	0.561659	17.681166
1024 ²	nonlinear head ($k=8$)	4.555479	4.555461	12.318828	36.325427
1024 ²	same-grid direct solve	0.000350	0.000000	1.034563	25.068664
1024 ²	coarse (16 ²) direct solve, interpolated	2.577910	2.578073	0.313116	24.364979

Table F.9: Neural operators on the shared Burgers data: capacity, epochs, whether the arm was still improving at its budget, worst and median error on the matched eight-case cohort and on the 32 held-out cases. Accuracy is comparable across jobs; timing never is, and none is shown. Single seed, one mesh, one 3000 s budget per capacity.

arm	family	params	epochs	still improving	8-case worst %	val-32 worst %	val-32 median %	job
same_ntle-2_dt005	FOM	—	—	—	0.9978	—	—	3702709
same_ntle-6_dt005	FOM	—	—	—	1.3811	—	—	3702709
same_ntle-4_dt005	FOM	—	—	—	1.3811	—	—	3702709
unet-small	U-Net	4,368,389	2327	yes	1.4712	5.5937	1.3963	3780138
unet-medium	U-Net	7,763,461	1963	yes	1.5189	3.9622	1.3169	3780138
tsol-refine	Transolver	3,108,240	1628	no	1.5224	9.3183	1.3591	3780139
unet-refine	U-Net	7,763,461	1962	yes	1.7110	7.5176	1.0806	3780138
same_ntle-2_dt01	FOM	—	—	—	1.8642	—	—	3702709
rom	ROM	—	—	—	1.8671	—	—	3702709
coarse_half_dt005	FOM	—	—	—	1.8993	—	—	3702709
unet-large	U-Net	17,462,021	908	yes	2.3176	4.2340	1.3642	3780138
fno-large	FNO	17,877,317	692	yes	2.4829	6.3825	1.8054	3710846
tsol-small	Transolver	3,108,240	1628	yes	2.5273	6.8213	1.4522	3780139
tsol-medium	Transolver	6,952,208	586	yes	2.8498	9.5338	2.3846	3780139
fno-medium	FNO	5,780,213	1198	yes	2.8882	6.0423	1.7638	3710846
fno-small	FNO	1,193,125	1741	yes	2.9350	7.4812	1.9552	3710846
tsol-large	Transolver	12,322,960	882	yes	3.0988	6.3953	3.0690	3780139
fno-refine	FNO	17,877,317	688	yes	3.2660	7.1747	1.8810	3710846
coarse_quarter_dt01	FOM	—	—	—	3.6398	—	—	3702709

Same-grid, reference, physical error

Error against the converged full-order solve on the same mesh (isolates the reduction); against a much finer solve (includes the mesh's own discretisation error); the cell's declared accuracy metric.

Three layers: bank floor, best-found, solved

Error of the best linear projection onto the bank (what nothing at inference can beat); the smallest error any point of the augmented manifold attains, found by a multistart oracle (an upper bound on the true minimum); the error the deployed iteration returns.

Stationary, budget exit, early-stopped, converged, strict

The solve met the normalised-gradient tolerance; it hit its iteration cap; any output produced under a cap or stall; the pre-registered completion rule that also accepts an attained initial fit whose residual is at round-off; the earlier rule that does not.

Admissible

Eligible for a frontier or a ratio: converged under the completion rule as written, with stationarity 10^{-6} at every step or a residual-rule exit, and for a quadrature subject a rule that passes its bar. Arms run at the loose tolerance 10^{-3} are reported everywhere but are not admissible.

Non-dominated, frontier, error span, cost span

A point nothing else beats on both axes; the set of such points; ratio of largest to smallest error among a ladder's non-dominated converged points; the same for cost. The tunability bar: monotone, at least three non-dominated points, at least $2\times$ on both spans, nothing early-stopped.

Same-allocation, same-job, cross-job

Two measurements inside one cluster allocation on one GPU, which may be divided; two from different allocations, which may not.

Table F.10: The correction ladder at 256^2 in one allocation (job 3789570, NVIDIA A100 80GB PCIe, checkpoint incumbent (gate checkpoint_unchanged; hash in T2, tuning row)), tolerance 10^{-6} : dense advection beside two quadrature rule sets, the rules the b-eqtop ladder used and the rules its replication selected (§6.3), with each rule’s status from that replication; worst error over evolved and over all times (%), median GPU ms, and the $t=0$ compression bounding the all-times metric. “Certified in one draw” is a single passing draw, not a certified construction. Every subject of the job, at both tolerances, is in Table F.13.

q	M	dense: evolved %	all %	ms	EQ, b-eqtop ladder rules: evolved %	all %	ms	rule status	EQ, replication-selected rules: evolved %	all %	ms	rule status	$t=0$ %
0	64	1.8890	2.5629	283.9	1.8891	2.5629	59.5	confirmed (3 of 3 re-draws)	1.8891	2.5629	58.6	confirmed (3 of 3 re-draws)	2.5629
16	128	1.3985	2.4806	360.8	1.4270	2.4806	81.7	confirmed (3 of 3 re-draws)	1.4270	2.4806	81.1	confirmed (3 of 3 re-draws)	2.4806
32	192	1.2336	2.3534	434.7	1.2493	2.3534	96.4	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.8	confirmed (2 of 2 re-draws)	2.3534
64	320	1.0843	2.1489	621.1	1.2275	2.1489	119.7	marginal (2 of 6 draws pass)	1.0840	2.1489	148.9	confirmed (2 of 2 re-draws)	2.1489
128	576	0.8930	1.8116	1186.9	0.8925	1.8116	246.5	marginal (4 of 5 draws pass)	0.8931	1.8116	273.5	single-draw	1.8116
256	1088	0.5194	0.9053	3939.8	1.0361	1.0361	697.6	marginal (1 of 5 draws pass)	0.5129	0.9053	746.0	single-draw	0.9053

Table F.11: The correction ladder at 1024^2 in one allocation (job 3789572, NVIDIA H200, same checkpoint), tolerance 10^{-6} : dense advection beside the quadrature rules transferred from 256^2 (rule status as at 256^2 ; the $q \geq 64$ rules miss the primary bar here, §6.1). Milliseconds are not comparable with Table F.10: different GPU, different job.

q	M	dense: evolved %	all %	ms	EQ, ladder rules transferred from 256^2 : evolved %	all %	ms	rule status	$t=0$ %
0	64	2.2886	3.8562	1551.7	2.2913	3.8562	40.3	transferred, primary at 1024^2 ; source confirmed (3 of 3 re-draws)	3.8562
16	128	1.5398	3.8071	1949.1	1.5273	3.8071	59.6	transferred, primary at 1024^2 ; source confirmed (3 of 3 re-draws)	3.8071
32	192	1.4482	3.7140	2382.9	1.4487	3.7140	72.0	transferred, primary at 1024^2 ; source confirmed (2 of 2 re-draws)	3.7140
64	320	1.2739	3.5773	3565.6	1.2762	3.5773	84.0	transferred, not primary at 1024^2 ; source marginal (2 of 6 draws pass)	3.5773
128	576	1.0444	3.2858	6895.2	1.0442	3.2858	163.7	transferred, not primary at 1024^2 ; source marginal (4 of 5 draws pass)	3.2858
256	1088	0.5861	2.7974	22053.9	1.8664	2.7974	502.2	transferred, not primary at 1024^2 ; source marginal (1 of 5 draws pass)	2.7974

Development cohort, held-out, sealed cohort

Cases opened and used for diagnosis; cases never used for selection; cases to be opened once, at the end, with every choice frozen (unopened for every cell here).

Pre-registered

Declared in a design document before the job ran; every bar, gate and falsification clause in this paper was.

POD-LSPG, k' , DST, CG, Newton

The classical linear reduced model on a snapshot basis of rank k' solved through the same weak objective; the direct discrete sine transform solve, exact for separable constant-coefficient operators on a rectangle; conjugate gradients; the full-order nonlinear iteration for Burgers.

FNO, U-Net, Transolver

Three neural-operator families. Each is fixed once trained; its evaluation grid is a cost–accuracy control of its own (Table F.31), but one that does not change what the model can represent.

Checkpoint, frozen, incumbent

Saved network weights; unchanged for every result; the one Burgers checkpoint every cell shares.

Provisional, pending

A number that is real but rests on one draw, one seed or an unfinished replication; a placeholder for a run that has not landed.

Table F.12: The correction ladder at 512^2 in one allocation (job 3805065, NVIDIA A100 80GB PCIe, same checkpoint and GPU model as 256^2), tolerance 10^{-6} : dense advection beside both 256^2 rule sets transferred to this mesh (each transfer refits on its own fit-state draw); the status names whether the transfer passes the primary bar here and its source rule’s verdict at 256^2 . Milliseconds are not comparable with Table F.10: a different job.

ϵ	M	dense evolved %	all %	ms	EQ ladder rules transferred from 256^2	evolved %	all %	ms	rule status	EQ replication-selected rules transferred from 256^2	evolved %	all %	ms	rule status	$t \pm 0.5$
0	64	2.1376	3.2836	923.2		2.1400	3.2836	58.0	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)		2.1378	3.2836	58.1	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)	3.2836
18	128	1.5071	3.2248	1190.8		1.4014	3.2248	79.6	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)		1.4037	3.2248	79.2	transferred, primary at 512^2 , source confirmed (3 of 3 re-draws)	3.2248
32	192	1.3606	3.1172	1862.4		1.3072	3.1172	96.7	transferred, primary at 512^2 , source confirmed (2 of 2 re-draws)		1.4085	3.1172	93.9	transferred, not primary at 512^2 , source confirmed (2 of 2 re-draws)	3.1172
64	320	1.2015	2.9565	2231.7		1.2485	2.9565	120.8	transferred, not primary at 512^2 , source marginal (5 of 6 draws pass)		1.1994	2.9565	150.2	transferred, not primary at 512^2 , source confirmed (2 of 2 re-draws)	2.9565
128	576	0.8900	2.6066	4841.3		0.8084	2.6066	242.0	transferred, not primary at 512^2 , source marginal (1 of 6 draws pass)		0.9001	2.6066	262.4	transferred, primary at 512^2 , source single-draw	2.6066
256	1088	0.5625	2.0195	16219.9		1.8284	2.0195	686.6	transferred, not primary at 512^2 , source marginal (1 of 5 draws pass)		0.5510	2.0195	783.3	transferred, primary at 512^2 , source single-draw	2.0195

Table F.13: Every timed subject of the 256^2 same-allocation panel (job 3789570). “conv. (§5)” is the pre-registered completion rule as written — stationarity 10^{-6} at every step, or a residual-rule exit, with an attained initial fit accepted; “strict” is the earlier rule that does not accept it, tested against the arm’s own tolerance (so a 10^{-3} arm can be strict and still not converged under §5); “adm. (§5)” is admissibility for a frontier or a ratio (converged under that rule, and for EQ subjects a rule passing a bar). “adm. (own tol.)” is the superseded flag: it tested each arm against its own tolerance and so admitted the 12 arms run at 10^{-3} , which the rule as written does not (0 of 12 pass it). Those arms are kept here with all their numbers but are not admissible, which moves the admissible reduced count at this mesh from 39 to 27. The correction is the panel pre-registration’s §A13, and every panel number in this paper is read from that lane at commit 25434a27.

subject	family	q/k'	M	quad.	rule	evolved %	all %	$t=0$ %	vs ref %	device ms	complete-spary ms	conv. (§5)	strict	adm. (§5)	adm. (own tol.)
q0_M256_dense_glem06	rom	0	256	dense	none	1.2710	2.5629	2.5629	4.0637	341.5	343.4	yes	yes	yes	yes
q0_M64_dense_glem06	rom	0	64	dense	none	1.8890	2.5629	2.5629	4.5575	283.9	285.9	yes	yes	yes	yes
q0_M64_eqcert_g0p001	rom	0	64	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.8898	2.5628	2.5628	4.5521	43.6	45.8	no	yes	no	yes
q0_M64_eqcert_glem06	rom	0	64	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.8891	2.5629	2.5629	4.5529	59.5	61.7	yes	yes	yes	yes
q0_M64_eqtop_g0p001	rom	0	64	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.8898	2.5628	2.5628	4.5521	43.4	45.5	no	yes	no	yes
q0_M64_eqtop_glem06	rom	0	64	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.8891	2.5629	2.5629	4.5529	58.6	60.6	yes	yes	yes	yes
q16_M128_dense_glem06	rom	16	128	dense	none	1.3985	2.4806	2.4806	4.1065	360.8	362.8	yes	yes	yes	yes
q16_M128_eqcert_g0p001	rom	16	128	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.4272	2.4806	2.4806	4.1136	63.3	65.3	no	yes	no	yes
q16_M128_eqcert_glem06	rom	16	128	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.4270	2.4806	2.4806	4.1145	81.7	83.8	yes	yes	yes	yes
q16_M128_eqtop_g0p001	rom	16	128	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.4272	2.4806	2.4806	4.1136	62.1	64.2	no	yes	no	yes
q16_M128_eqtop_glem06	rom	16	128	eq	eqtop m=1024, confirmed (3 of 3 re-draws)	1.4270	2.4806	2.4806	4.1145	81.1	83.3	yes	yes	yes	yes
q32_M192_dense_glem06	rom	32	192	dense	none	1.2336	2.3534	2.3534	4.0814	434.7	437.0	yes	yes	yes	yes
q32_M192_eqcert_g0p001	rom	32	192	eq	eqcert m=1024, confirmed (2 of 2 re-draws)	1.2497	2.3534	2.3534	4.0805	73.7	75.7	no	yes	no	yes
q32_M192_eqcert_glem06	rom	32	192	eq	eqcert m=1024, confirmed (2 of 2 re-draws)	1.2493	2.3534	2.3534	4.0818	96.4	98.4	yes	yes	yes	yes
q32_M192_eqtop_g0p001	rom	32	192	eq	eqtop m=1024, confirmed (2 of 2 re-draws)	1.2497	2.3534	2.3534	4.0805	74.1	76.2	no	yes	no	yes
q32_M192_eqtop_glem06	rom	32	192	eq	eqtop m=1024, confirmed (2 of 2 re-draws)	1.2493	2.3534	2.3534	4.0818	97.8	99.8	yes	yes	yes	yes
q64_M320_dense_glem06	rom	64	320	dense	none	1.0843	2.1489	2.1489	4.1013	621.1	623.0	yes	yes	yes	yes
q64_M320_eqcert_g0p001	rom	64	320	eq	eqcert m=1024, marginal (2 of 6 draws pass)	1.2278	2.1489	2.1489	4.0836	90.8	92.6	no	yes	no	yes
q64_M320_eqcert_glem06	rom	64	320	eq	eqcert m=1024, marginal (2 of 6 draws pass)	1.2275	2.1489	2.1489	4.0855	119.7	121.8	yes	yes	yes	yes
q64_M320_eqtop_g0p001	rom	64	320	eq	eqtop m=2048, confirmed (2 of 2 re-draws)	1.0841	2.1489	2.1489	4.1008	113.9	116.0	no	yes	no	yes
q64_M320_eqtop_glem06	rom	64	320	eq	eqtop m=2048, confirmed (2 of 2 re-draws)	1.0840	2.1489	2.1489	4.1027	148.9	151.0	yes	yes	yes	yes
q128_M576_dense_glem06	rom	128	576	dense	none	0.8930	1.8116	1.8116	4.0797	1186.9	1189.0	yes	yes	yes	yes
q128_M576_eqcert_g0p001	rom	128	576	eq	eqcert m=2048, marginal (4 of 5 draws pass)	0.8926	1.8116	1.8116	4.0791	189.3	191.5	no	yes	no	yes
q128_M576_eqcert_glem06	rom	128	576	eq	eqcert m=2048, marginal (4 of 5 draws pass)	0.8925	1.8116	1.8116	4.0809	246.5	248.6	yes	yes	yes	yes
q128_M576_eqtop_g0p001	rom	128	576	eq	eqtop m=2319, single-draw	0.8932	1.8116	1.8116	4.0787	205.1	207.2	no	yes	no	yes
q128_M576_eqtop_glem06	rom	128	576	eq	eqtop m=2319, single-draw	0.8931	1.8116	1.8116	4.0806	273.5	275.3	yes	yes	yes	yes
q256_M1088_dense_glem06	rom	256	1088	dense	none	0.5194	0.9053	0.9053	4.0391	3939.8	3942.0	yes	yes	yes	yes
q256_M1088_eqcert_g0p001	rom	256	1088	eq	eqcert m=2048, marginal (1 of 5 draws pass)	1.0324	1.0324	0.9053	4.0322	429.6	431.6	no	yes	no	yes
q256_M1088_eqcert_glem06	rom	256	1088	eq	eqcert m=2048, marginal (1 of 5 draws pass)	1.0361	1.0361	0.9053	4.0332	697.6	699.7	yes	yes	yes	yes
q256_M1088_eqtop_g0p001	rom	256	1088	eq	eqtop m=2560, single-draw	0.5129	0.9053	0.9053	4.0355	466.4	468.6	no	yes	no	yes
q256_M1088_eqtop_glem06	rom	256	1088	eq	eqtop m=2560, single-draw	0.5129	0.9053	0.9053	4.0365	746.0	748.4	yes	yes	yes	yes
q0_M64_eqcert_glem06_FastL4	fast	0	64	eq	eqcert m=1024, confirmed (3 of 3 re-draws)	1.8891	2.5629	2.5629	4.5529	40.4	42.5	yes	yes	yes	yes
pod16_M64_dense	pod	16	64	dense	none	28.7250	61.6503	61.6503	61.6503	46.8	48.8	yes	yes	yes	yes
pod32_M128_dense	pod	32	128	dense	none	18.7995	47.0681	47.0681	47.0681	81.0	82.9	yes	yes	yes	yes
pod64_M256_dense	pod	64	256	dense	none	7.0835	19.8156	19.8156	19.8156	145.7	147.7	yes	yes	yes	yes
pod128_M512_dense	pod	128	512	dense	none	1.9464	10.1198	10.1198	10.1198	331.6	333.5	yes	yes	yes	yes
pod256_M1024_dense	pod	256	1024	dense	none	0.7109	3.7698	3.7698	4.0362	898.8	901.2	yes	yes	yes	yes
pod512_M2048_dense	pod	512	2048	dense	none	0.2184	0.6125	0.6125	4.0266	2790.8	2794.3	yes	yes	yes	yes
free512_M1024_dense	free	512	1024	dense	none	0.4471	0.6027	0.6027	4.0423	2439.6	2441.9	yes	yes	yes	yes
fno-large	fno	—	—	—	—	7.4164	7.4164	0.0000	5.7495	7.2	7.4	—	—	—	—
dense_1ight	fno	—	—	—	—	0.0000	0.0000	0.0000	4.0265	65.9	65.8	—	—	—	—
fft_1ight	fno	—	—	—	—	0.0000	0.0000	0.0000	4.0265	90.4	92.5	—	—	—	—
ntle-2_at005	fno	—	—	—	—	3.7127	3.7127	0.0000	2.4737	15.7	17.7	—	—	—	—
ntle-2_at01	fno	—	—	—	—	3.1999	3.1999	0.0000	3.6168	9.0	11.0	—	—	—	—
ntle-3_at005	fno	—	—	—	—	0.0489	0.0489	0.0000	4.0399	31.8	34.2	—	—	—	—
ntle-3_at01	fno	—	—	—	—	1.5179	1.5179	0.0000	5.1761	19.7	21.7	—	—	—	—
ntle-4_at005	fno	—	—	—	—	0.0338	0.0338	0.0000	4.0320	37.0	39.0	—	—	—	—
ntle-4_at01	fno	—	—	—	—	1.5109	1.5109	0.0000	5.1552	27.8	30.0	—	—	—	—

Table F.14: Every timed subject of the 1024² same-allocation panel (job 3789572, H200). Columns as in Table F.13; milliseconds are comparable only within this table. The §5 rule takes the admissible reduced count from 20 to 16 and the reduced subjects on the (GPU ms, worst evolved) frontier from 5 to 3 (q0_M64_eqxfer_g1em06, q16_M128_eqxfer_g1em06, q32_M192_eqxfer_g1em06).

subject	family	q / k'	M	quad.	rule	evolved %	all %	t=0 %	vs ref %	device ms	complete-query ms	conv. (§5)	strict	adm. (§5)	adm. (own tol.)
q0_M64_dense_g1em06	rom	0	64	dense	none	2.2886	3.8562	3.8562	3.8562	1551.7	1562.4	yes	yes	yes	yes
q0_M64_eqxfer_g0p001	rom	0	64	eq	eqxfer m=934, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	2.2925	3.8562	3.8562	3.8562	32.2	44.4	no	yes	no	yes
q0_M64_eqxfer_g1em06	rom	0	64	eq	eqxfer m=934, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	2.2913	3.8562	3.8562	3.8562	40.3	51.4	yes	yes	yes	yes
q16_M128_dense_g1em06	rom	16	128	dense	none	1.5398	3.8071	3.8071	3.8071	1949.1	1960.6	yes	yes	yes	yes
q16_M128_eqxfer_g0p001	rom	16	128	eq	eqxfer m=918, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	1.5277	3.8071	3.8071	3.8071	48.4	60.7	no	yes	no	yes
q16_M128_eqxfer_g1em06	rom	16	128	eq	eqxfer m=918, transferred, primary at 1024 ² ; source confirmed (3 of 3 re-draws)	1.5273	3.8071	3.8071	3.8071	59.6	70.5	yes	yes	yes	yes
q32_M192_dense_g1em06	rom	32	192	dense	none	1.4482	3.7140	3.7140	3.7140	2382.9	2393.5	yes	yes	yes	yes
q32_M192_eqxfer_g0p001	rom	32	192	eq	eqxfer m=972, transferred, primary at 1024 ² ; source confirmed (2 of 2 re-draws)	1.4488	3.7140	3.7140	3.7140	57.6	69.2	no	yes	no	yes
q32_M192_eqxfer_g1em06	rom	32	192	eq	eqxfer m=972, transferred, primary at 1024 ² ; source confirmed (2 of 2 re-draws)	1.4487	3.7140	3.7140	3.7140	72.0	83.0	yes	yes	yes	yes
q64_M320_dense_g1em06	rom	64	320	dense	none	1.2739	3.5773	3.5773	3.5773	3565.6	3581.8	yes	yes	yes	yes
q64_M320_eqxfer_g0p001	rom	64	320	eq	eqxfer m=942, transferred, not primary at 1024 ² ; source marginal (2 of 6 draws pass)	1.2763	3.5773	3.5773	3.5773	64.4	76.5	no	yes	no	yes
q64_M320_eqxfer_g1em06	rom	64	320	eq	eqxfer m=942, transferred, not primary at 1024 ² ; source marginal (2 of 6 draws pass)	1.2762	3.5773	3.5773	3.5773	84.0	94.5	yes	yes	no	no
q128_M576_dense_g1em06	rom	128	576	dense	none	1.0444	3.2858	3.2858	3.2858	6895.2	6909.1	yes	yes	yes	yes
q128_M576_eqxfer_g0p001	rom	128	576	eq	eqxfer m=1713, transferred, not primary at 1024 ² ; source marginal (4 of 5 draws pass)	1.0443	3.2858	3.2858	3.2858	128.3	139.2	no	yes	no	yes
q128_M576_eqxfer_g1em06	rom	128	576	eq	eqxfer m=1713, transferred, not primary at 1024 ² ; source marginal (4 of 5 draws pass)	1.0442	3.2858	3.2858	3.2858	163.7	175.4	yes	yes	yes	yes
q256_M1088_dense_g1em06	rom	256	1088	dense	none	0.5861	2.7974	2.7974	2.7974	22053.9	22066.3	yes	yes	yes	yes
q256_M1088_eqxfer_g0p001	rom	256	1088	eq	eqxfer m=1658, transferred, not primary at 1024 ² ; source marginal (1 of 5 draws pass)	1.8655	2.7974	2.7974	2.7974	277.2	290.1	no	yes	no	yes
q256_M1088_eqxfer_g1em06	rom	256	1088	eq	eqxfer m=1658, transferred, not primary at 1024 ² ; source marginal (1 of 5 draws pass)	1.8664	2.7974	2.7974	2.7974	502.2	515.1	yes	yes	no	no
pod16_M64_dense	pod	16	64	dense	none	29.2793	61.9285	61.9285	61.9285	198.2	209.6	yes	yes	yes	yes
pod32_M128_dense	pod	32	128	dense	none	19.4088	47.6410	47.6410	47.6410	391.2	402.5	yes	no	yes	yes
pod64_M256_dense	pod	64	256	dense	none	7.4077	20.5994	20.5994	20.5994	741.6	752.9	yes	no	yes	yes
pod128_M512_dense	pod	128	512	dense	none	2.1609	10.7339	10.7339	10.7339	1760.6	1773.0	yes	no	yes	yes
pod256_M1024_dense	pod	256	1024	dense	none	1.1017	4.1416	4.1416	4.1416	4897.6	4911.4	yes	no	yes	yes
free512_M1024_dense	free	512	1024	dense	none	0.5108	2.4466	2.4466	2.4466	13514.6	13529.0	yes	no	yes	yes
free-large	fno	—	—	—	—	6.2657	0.0000	0.0000	5.6472	58.9	71.0	—	—	—	—
fft-tight	fom	—	—	—	—	0.0000	0.0000	0.0000	2.1416	210.9	222.2	—	—	—	—
nt1e-2_dt005	fom	—	—	—	—	4.2628	4.2628	0.0000	2.3899	31.1	42.7	—	—	—	—
nt1e-2_dt01	fom	—	—	—	—	3.4582	3.4582	0.0000	2.8919	17.7	28.3	—	—	—	—
nt1e-4_dt005	fom	—	—	—	—	0.0343	0.0343	0.0000	2.1281	81.3	92.5	—	—	—	—
nt1e-4_dt01	fom	—	—	—	—	1.6287	1.6287	0.0000	3.7562	64.0	76.3	—	—	—	—

Table F.15: Every timed subject of the 512² same-allocation panel (job 3805065). Columns as in Table F.13; milliseconds are comparable only within this table. The \$5 rule takes the admissible reduced count from 31 to 23; nothing reduced is on the frontier under either flag.

subject	family	q / k'	M	quad.	rule	evolved %	all %	$t=0$ %	vs ref %	device ms	complete-query ms	conv. (\$)	strict	adm. (\$5)	adm. (own tol.)
q0_M256_dense_glem06	rom	0	256	dense	none	1.3761	3.2836	3.2836	3.2836	1175.7	1179.7	yes	yes	yes	yes
q0_M64_dense_glem06	rom	0	64	dense	none	2.1376	3.2836	3.2836	3.7578	923.2	927.3	yes	yes	yes	yes
q0_M64_eqtqpfer_g0p001	rom	0	64	eq	eqtqpfer $m=914$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1387	3.2836	3.2836	3.7448	43.8	47.5	no	yes	no	yes
q0_M64_eqtqpfer_glem06	rom	0	64	eq	eqtqpfer $m=914$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1378	3.2836	3.2836	3.7449	59.1	62.9	yes	yes	yes	yes
q0_M64_eqxfer_g0p001	rom	0	64	eq	eqxfer $m=922$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1409	3.2836	3.2836	3.7681	44.1	48.3	no	yes	no	yes
q0_M64_eqxfer_glem06	rom	0	64	eq	eqxfer $m=922$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1400	3.2836	3.2836	3.7681	58.0	61.9	yes	yes	yes	yes
q16_M128_dense_glem06	rom	16	128	dense	none	1.5071	3.2248	3.2248	3.2248	1190.8	1194.9	yes	yes	yes	yes
q16_M128_eqtqpfer_g0p001	rom	16	128	eq	eqtqpfer $m=879$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4538	3.2248	3.2248	3.2248	62.5	66.3	no	yes	no	yes
q16_M128_eqtqpfer_glem06	rom	16	128	eq	eqtqpfer $m=879$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4537	3.2248	3.2248	3.2248	79.2	83.1	yes	yes	yes	yes
q16_M128_eqxfer_g0p001	rom	16	128	eq	eqxfer $m=911$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4917	3.2248	3.2248	3.2248	62.5	66.7	no	yes	no	yes
q16_M128_eqxfer_glem06	rom	16	128	eq	eqxfer $m=911$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	1.4914	3.2248	3.2248	3.2248	79.6	83.7	yes	yes	yes	yes
q32_M192_dense_glem06	rom	32	192	dense	none	1.3666	3.1172	3.1172	3.1172	1502.4	1506.1	yes	yes	yes	yes
q32_M192_eqtqpfer_g0p001	rom	32	192	eq	eqtqpfer $m=848$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.4060	3.1172	3.1172	3.1172	72.1	75.9	no	yes	no	yes
q32_M192_eqtqpfer_glem06	rom	32	192	eq	eqtqpfer $m=848$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.4055	3.1172	3.1172	3.1172	93.9	97.5	yes	yes	yes	yes
q32_M192_eqxfer_g0p001	rom	32	192	eq	eqxfer $m=973$, transferred, primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.3673	3.1172	3.1172	3.1172	74.4	78.6	no	yes	no	yes
q32_M192_eqxfer_glem06	rom	32	192	eq	eqxfer $m=973$, transferred, primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.3672	3.1172	3.1172	3.1172	96.7	100.8	yes	yes	yes	yes
q64_M320_dense_glem06	rom	64	320	dense	none	1.2015	2.9565	2.9565	3.0133	2221.7	2225.7	yes	yes	yes	yes
q64_M320_eqtqpfer_g0p001	rom	64	320	eq	eqtqpfer $m=1898$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.1995	2.9565	2.9565	3.0056	115.1	119.1	no	yes	no	no
q64_M320_eqtqpfer_glem06	rom	64	320	eq	eqtqpfer $m=1898$, transferred, not primary at 512 ² ; source confirmed (2 of 2 re-draws)	1.1994	2.9565	2.9565	3.0069	150.2	154.0	yes	yes	no	no
q64_M320_eqxfer_g0p001	rom	64	320	eq	eqxfer $m=937$, transferred, not primary at 512 ² ; source marginal (2 of 6 draws pass)	1.2488	2.9565	2.9565	3.0135	89.8	93.7	no	yes	no	no
q64_M320_eqxfer_glem06	rom	64	320	eq	eqxfer $m=937$, transferred, not primary at 512 ² ; source marginal (2 of 6 draws pass)	1.2485	2.9565	2.9565	3.0149	120.8	125.1	yes	yes	no	no
q128_M576_dense_glem06	rom	128	576	dense	none	0.9890	2.6066	2.6066	2.9236	4541.3	4545.3	yes	yes	yes	yes
q128_M576_eqtqpfer_g0p001	rom	128	576	eq	eqtqpfer $m=2152$, transferred, primary at 512 ² ; source single-draw	0.9902	2.6066	2.6066	2.9224	200.1	204.1	no	yes	no	yes
q128_M576_eqtqpfer_glem06	rom	128	576	eq	eqtqpfer $m=2152$, transferred, primary at 512 ² ; source single-draw	0.9901	2.6066	2.6066	2.9237	262.4	266.3	yes	yes	yes	yes
q128_M576_eqxfer_g0p001	rom	128	576	eq	eqxfer $m=1644$, transferred, not primary at 512 ² ; source marginal (4 of 5 draws pass)	0.9895	2.6066	2.6066	2.9188	188.4	192.4	no	yes	no	no
q128_M576_eqxfer_glem06	rom	128	576	eq	eqxfer $m=1644$, transferred, not primary at 512 ² ; source marginal (4 of 5 draws pass)	0.9894	2.6066	2.6066	2.9200	242.0	245.9	yes	yes	no	no
q256_M1088_dense_glem06	rom	256	1088	dense	none	0.5625	2.0195	2.0195	2.7611	16219.9	16224.4	yes	yes	yes	yes
q256_M1088_eqtqpfer_g0p001	rom	256	1088	eq	eqtqpfer $m=2438$, transferred, primary at 512 ² ; source single-draw	0.5510	2.0195	2.0195	2.7581	486.3	490.4	no	yes	no	yes
q256_M1088_eqtqpfer_glem06	rom	256	1088	eq	eqtqpfer $m=2438$, transferred, primary at 512 ² ; source single-draw	0.5510	2.0195	2.0195	2.7602	783.3	787.8	yes	yes	yes	yes
q256_M1088_eqxfer_g0p001	rom	256	1088	eq	eqxfer $m=1663$, transferred, not primary at 512 ² ; source marginal (1 of 5 draws pass)	1.8179	2.0195	2.0195	2.7920	395.7	399.9	no	yes	no	no
q256_M1088_eqxfer_glem06	rom	256	1088	eq	eqxfer $m=1663$, transferred, not primary at 512 ² ; source marginal (1 of 5 draws pass)	1.8204	2.0195	2.0195	2.7969	686.6	690.7	yes	yes	no	no
q0_M64_eqxfer_glem06_Fast-L4	fast	0	64	eq	eqxfer $m=922$, transferred, primary at 512 ² ; source confirmed (3 of 3 re-draws)	2.1400	3.2836	3.2836	3.7681	40.5	44.4	yes	yes	yes	yes
pod16_M64_dense	pod	16	64	dense	none	29.0916	61.8330	61.8330	61.8330	127.5	131.5	yes	yes	yes	yes
pod32_M128_dense	pod	32	128	dense	none	19.1988	47.4440	47.4440	47.4440	247.8	251.7	yes	yes	yes	yes
pod64_M256_dense	pod	64	256	dense	none	7.2900	20.3198	20.3198	20.3198	488.5	492.5	yes	yes	yes	yes
pod128_M512_dense	pod	128	512	dense	none	2.0372	10.5346	10.5346	10.5346	1206.7	1210.7	yes	yes	yes	yes
pod256_M1024_dense	pod	256	1024	dense	none	0.9578	4.0010	4.0010	4.0010	3488.2	3492.8	yes	yes	yes	yes
pod512_M2048_dense	pod	512	2048	dense	none	0.3328	0.6579	0.6579	0.7299	11933.9	11941.1	yes	yes	yes	yes
free512_M1024_dense	free	512	1024	dense	none	0.4912	1.8046	1.8046	2.7232	10019.3	10024.2	yes	yes	yes	yes
fno-large	fno	—	—	—	—	6.6169	6.6169	0.0000	5.6812	22.5	23.2	—	—	—	—
dense_x_tight	fom	—	—	—	—	0.0000	0.0000	0.0000	2.7025	147.4	151.4	—	—	—	—
fft_x_tight	fom	—	—	—	—	0.0000	0.0000	0.0000	2.7025	157.7	161.8	—	—	—	—
ntile-2_at005	fom	—	—	—	—	4.0614	4.0614	0.0000	2.0638	24.4	28.3	—	—	—	—
ntile-2_at01	fom	—	—	—	—	3.3550	3.3550	0.0000	3.3046	13.9	17.8	—	—	—	—
ntile-3_at005	fom	—	—	—	—	0.0516	0.0516	0.0000	2.7116	52.9	56.5	—	—	—	—
ntile-3_at01	fom	—	—	—	—	1.5893	1.5893	0.0000	4.1814	33.7	37.7	—	—	—	—
ntile-4_at005	fom	—	—	—	—	0.0341	0.0341	0.0000	2.6975	61.2	65.4	—	—	—	—
ntile-4_at01	fom	—	—	—	—	1.5869	1.5869	0.0000	4.1611	48.4	52.3	—	—	—	—

Table F.16: Fixed- q sweeps in the test count M (dense; errors comparable across the three b-qxm jobs, costs not shown). The last column restricts the span to at least two tests per unknown.

q	M sweep	worst evolved %	monotone	span	best M	span, $M \geq 2(K+q)$
0	64, 128, 256, 512, 1088, 2048, 4096	1.8890 / 1.4695 / 1.2710 / 1.2662 / 1.2657 / 1.2652 / 1.2649	yes	1.49×	4096	1.49×
16	128, 192, 256, 512, 1088	1.3985 / 1.2690 / 1.2305 / 1.2204 / 1.2204	yes	1.15×	1088	1.15×
32	192, 256, 320, 384, 768, 1088	1.2336 / 1.1869 / 1.1717 / 1.1665 / 1.1634 / 1.1629	yes	1.06×	1088	1.06×
64	128, 256, 320, 512, 576, 640, 1088, 1280, 2048, 4096	1.8032 / 1.1255 / 1.0843 / 1.0647 / 1.0633 / 1.0619 / 1.0593 / 1.0588 / 1.0594 / 1.0620	no	1.70×	1280	1.06×
128	256, 576, 1088, 1152	1.0418 / 0.8930 / 0.8711 / 0.8689	yes	1.20×	1152	1.03×
256	544, 1088, 2176, 3264, 4352, 6528	0.7566 / 0.5194 / 0.3736 / 0.3564 / 0.3523 / 0.3510	yes	2.16×	6528	2.16×

Table F.17: Head ablation, Burgers 256² (job 3711424, NVIDIA A100 80GB PCIe). Same-grid error against the converged full-order solve; “vs ref” against the 4096² reference.

arm	dim.	bank floor %	best-found %	solved same-grid %	vs ref %	device ms	stationary
(a) neural head, EQ	16	0.3918	2.5447	2.5629	4.5546	47.6	yes
(a) neural head, dense	16	0.3918	2.5447	2.5629	4.5575	281.7	yes
(b) linear map (fit to head outputs)	16	0.3918	56.9296	56.9296	56.9296	19.9	yes
(b) linear map (fit to truth)	16	0.3918	61.5226	61.5226	61.5226	19.7	yes
(c) quadratic map	16	0.3918	31.8276	31.8276	31.8276	28.0	yes
(e) POD-LSPG $k'=16$	16	61.6503	61.6503	61.6503	61.6503	46.0	yes
(e) POD-LSPG $k'=32$	32	47.0681	47.0681	47.0681	47.0681	80.7	yes
(e) POD-LSPG $k'=64$	64	19.8147	19.8147	19.8156	19.8156	144.1	no
(e) POD-LSPG $k'=128$	128	10.1189	10.1189	10.1198	10.1198	328.3	no
(d) unrestricted bank ($R=512$)	512	0.3918	0.3918	0.6027	4.0423	2417.6	no
FOM Newton, loose	—	—	—	3.7127	2.4737	15.4	—
FOM Newton, tight (reference)	—	—	—	0.0000	4.0265	88.7	—

Table F.18: Head ablation, Poisson 1024², $R = 128$ (job 3711736, NVIDIA A100 80GB PCIe).

arm	dim.	bank floor %	best-found %	solved %	query ms
(a) neural head	16	2.3148	6.0926	6.0927	5.843
(a+) head + 32 eliminated corrections	16	—	—	4.6670	7.317
(b) linear map (head outputs)	16	2.3148	17.2966	17.2966	5.163
(b) linear map (truth)	16	2.3148	21.1648	21.1648	5.329
(c) quadratic map	16	2.3148	16.6302	16.6303	5.774
(e) POD-LSPG $k'=8$	8	22.3196	22.3196	22.3196	4.633
(e) POD-LSPG $k'=16$	16	20.0548	20.0548	20.0548	4.659
(e) POD-LSPG $k'=32$	32	11.8589	11.8589	11.8589	5.050
(e) POD-LSPG $k'=64$	64	7.7397	7.7397	7.7398	5.170
(e) POD-LSPG $k'=128$	128	4.3213	4.3213	4.3452	5.930
(d) unrestricted bank ($R=128$)	128	2.3148	2.3148	2.3274	5.995
direct DST	—	—	—	0.0007	4.341

Table F.19: Solver knobs at a fixed Burgers checkpoint on 32 held-out cases (job 3712269, NVIDIA A100 80GB PCIe, checkpoint 18f0266ae6f0. . .). Loosening the tolerance from 10^{-8} to 10^{-3} removes 35.6 % of the query for a change from 6.712 to 6.701 % worst error; a cap of 2 is a cliff (90.3 %). The full-order control at tolerance 10^{-4} beats the best ROM setting on both axes.

setting	worst % (32 held-out)	device ms	early-stopped
EQ $m=256$, tol 10^{-6} , cap 180	7.245	51.0	0/96
EQ $m=512$, tol 10^{-8}	6.712	64.7	0/96
EQ $m=512$, tol 10^{-3}	6.701	41.6	0/96
EQ $m=512$, cap 2 (early-stopped)	90.324	28.4	96/96
FOM Newton 10^{-2} , $\Delta t=0.01$	6.807	9.9	0/96
FOM Newton 10^{-2} , $\Delta t=0.005$	35.357	17.4	0/96
FOM Newton 10^{-4} , $\Delta t=0.005$	6.171	22.4	0/96
FOM Newton 10^{-6} , $\Delta t=0.005$	6.172	88.3	0/96
FOM 128^2 , $\Delta t=0.005$	9.849	21.2	0/96
FOM 64^2 , $\Delta t=0.01$	16.095	14.0	0/96

Table F.20: Cold-start iteration-cap sweep on Poisson at 1024 intervals (local GB10 diagnostic; only ratios transfer). From the nearest training code the converged answer is reached at a small cap; from $z = 0$ the cap traces a monotone curve that looks like an accuracy–cost frontier, every point of which is dominated by starting well.

start	LM cap	worst %	median %	median iters	ms (GB10, ratios only)
nearest	1	13.55	4.29	1	20.18
nearest	2	6.80	1.29	2	25.68
nearest	4	6.09	1.24	4	33.83
nearest	6	6.09	1.24	5	40.58
nearest	8	6.09	1.24	5	38.67
nearest	10	6.09	1.24	5	38.23
nearest	12	6.09	1.24	5	38.92
nearest	20	6.09	1.24	5	38.47
nearest	300	6.09	1.24	5	70.17
zero	1	91.88	74.88	1	20.60
zero	2	91.88	74.88	2	25.46
zero	4	91.88	74.88	4	33.43
zero	6	91.88	54.37	6	45.09
zero	8	30.03	5.96	8	52.23
zero	10	8.19	2.10	10	61.98
zero	12	6.09	1.46	12	71.31
zero	20	6.09	1.24	14	85.48
zero	300	6.09	1.24	14	80.89

Table F.21: Cheapest rule passing each bar per rung in its own draw, with its fit-state count, and the parent lane’s rule re-scored on the same held-out states. Jobs bet101 = 3780164 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet201 = 3780165 (NVIDIA A100 80GB PCIe, commit ace8936e42b3...); bet301 = 3783811 (NVIDIA A100-PCIe-40GB, commit b2844c607290...). Construction status is in Table F.25.

q	cheapest rule passing the primary bar (its draw)				cheapest rule passing the tight bar (its draw)				best parent-lane rule, re-scored: primary?
0	this_job/std,	$m=512$,	64	states,	this_job/rhow64,	$m=622$,	64	states,	$m=1024$, $\rho_{\max}=0.0153$, yes
	$\rho_{\max}=0.0641$				$\rho_{\max}=0.0491$				
16	this_job/std,	$m=512$,	64	states,	this_job/std,	$m=1024$,	64	states,	$m=2048$, $\rho_{\max}=0.0452$, yes
	$\rho_{\max}=0.0879$				$\rho_{\max}=0.0118$				
32	qrg304/reachable,	$m=1024$,	42	states,	qrg304/reachable,	$m=1024$,	42	states,	$m=1024$, $\rho_{\max}=0.0533$, yes
	$\rho_{\max}=0.0533$				$\rho_{\max}=0.0533$				
64	qrg304/reachable,	$m=1024$,	25	states,	qrg304/reachable,	$m=1024$,	25	states,	$m=1024$, $\rho_{\max}=0.0531$, yes
	$\rho_{\max}=0.0531$				$\rho_{\max}=0.0531$				
128	this_job/fs64,	$m=2048$,	64	states,	this_job/rhow64,	$m=2048$,	64	states,	$m=2048$, $\rho_{\max}=0.1908$, no
	$\rho_{\max}=0.0669$				$\rho_{\max}=0.0472$				
256	this_job/fs64,	$m=2048$,	64	states,	—				$m=2048$, $\rho_{\max}=0.1678$, no
	$\rho_{\max}=0.1074$								

Table F.22: Every quadrature rule at $q \leq 32$: NNLS fit residual beside the held-out ρ , so the anti-correlation stays visible. Bars: primary $\rho_{\max} \leq 0.116$, tight $\rho_{\max} \leq 0.06$ (source: b-eqtop/DESIGN.md §5 (byte-identical from commit 33e8bcd, 2026-09-17 01:11, through the pinned 8542c604); origin q-ridge/DESIGN.md amendment A3, commit a1822ae6 (2026-09-16); every b-eqtop job post-dates it). Continued in Table F.23.

q	fit arm	m	population	NNLS rel. fit	$\rho_{\max}$	ρ_{95}	primary	job
0	reachable	1024	qrg304:reachable	4.58e-05	0.0153	0.0088	yes	3780164
0	reachable	1024	qrg304:reachable	4.58e-05	0.0153	0.0088	yes	3780165
0	reachable	1024	qrg304:reachable	4.58e-05	0.0153	0.0088	yes	3783811
0	reachable	1521	qrg304:reachable	7.83e-06	0.0261	0.0083	yes	3780164
0	reachable	1521	qrg304:reachable	7.83e-06	0.0261	0.0083	yes	3780165
0	reachable	1521	qrg304:reachable	7.83e-06	0.0261	0.0083	yes	3783811
0	static	256	qrg304:static	6.85e-03	0.2407	0.2073	no	3780164
0	static	256	qrg304:static	6.85e-03	0.2407	0.2073	no	3780165
0	static	256	qrg304:static	6.85e-03	0.2407	0.2073	no	3783811
0	fs64	1024	this_job:reachable	4.70e-05	0.0424	0.0154	yes	3780165
0	fs64	1546	this_job:reachable	8.92e-06	0.0177	0.0091	yes	3780165
0	rhow64	609	this_job:reachable	2.24e-04	0.0686	0.0550	yes	3780165
0	rhow64	622	this_job:reachable	2.24e-04	0.0491	0.0328	yes	3780165
0	std	256	this_job:reachable	4.03e-03	0.4018	0.3674	no	3780165
0	std	512	this_job:reachable	4.07e-04	0.0641	0.0545	yes	3780165
0	std	1024	this_job:reachable	4.35e-05	0.0255	0.0087	yes	3780165
0	std	1526	this_job:reachable	9.17e-06	0.0135	0.0032	yes	3780165
0	std	1528	this_job:reachable	8.90e-06	0.0124	0.0038	yes	3780165
0	std	1530	this_job:reachable	9.07e-06	0.0128	0.0040	yes	3780165
16	reachable	1024	qrg304:reachable	1.90e-04	0.0935	0.0273	yes	3780164
16	reachable	1024	qrg304:reachable	1.90e-04	0.0935	0.0273	yes	3780165
16	reachable	1024	qrg304:reachable	1.90e-04	0.0935	0.0273	yes	3783811
16	reachable	2048	qrg304:reachable	1.92e-05	0.0452	0.0041	yes	3780164
16	reachable	2048	qrg304:reachable	1.92e-05	0.0452	0.0041	yes	3780165
16	reachable	2048	qrg304:reachable	1.92e-05	0.0452	0.0041	yes	3783811
16	static	512	qrg304:static	1.54e-03	0.1279	0.0523	no	3780164
16	static	512	qrg304:static	1.54e-03	0.1279	0.0523	no	3780165
16	static	512	qrg304:static	1.54e-03	0.1279	0.0523	no	3783811
16	fs64	1024	this_job:reachable	1.79e-04	0.0751	0.0232	yes	3780165
16	fs64	2048	this_job:reachable	1.59e-05	0.0148	0.0043	yes	3780165
16	rhow64	946	this_job:reachable	2.30e-04	0.0618	0.0174	yes	3780165
16	rhow64	972	this_job:reachable	2.33e-04	0.0603	0.0098	yes	3780165
16	std	256	this_job:reachable	3.09e-02	0.4133	0.3926	no	3780165
16	std	512	this_job:reachable	1.99e-03	0.0879	0.0723	yes	3780165
16	std	1024	this_job:reachable	2.42e-04	0.0118	0.0108	yes	3780165
16	std	2048	this_job:reachable	2.31e-05	0.0070	0.0031	yes	3780165
16	std	2556	this_job:reachable	8.89e-06	0.0021	0.0015	yes	3780165
16	std	2581	this_job:reachable	8.67e-06	0.0020	0.0014	yes	3780165
32	reachable	1024	qrg304:reachable	2.89e-04	0.0533	0.0226	yes	3780164
32	reachable	1024	qrg304:reachable	2.89e-04	0.0533	0.0226	yes	3780165
32	reachable	1024	qrg304:reachable	2.89e-04	0.0533	0.0226	yes	3783811
32	reachable	2048	qrg304:reachable	2.66e-05	0.0668	0.0120	yes	3780164
32	reachable	2048	qrg304:reachable	2.66e-05	0.0668	0.0120	yes	3780165
32	reachable	2048	qrg304:reachable	2.66e-05	0.0668	0.0120	yes	3783811
32	static	768	qrg304:static	6.27e-04	0.1482	0.0214	no	3780164
32	static	768	qrg304:static	6.27e-04	0.1482	0.0214	no	3780165
32	static	768	qrg304:static	6.27e-04	0.1482	0.0214	no	3783811
32	fs64	1024	this_job:reachable	3.80e-04	0.0696	0.0324	yes	3780165
32	fs64	2048	this_job:reachable	4.21e-05	0.0417	0.0036	yes	3780165
32	rhow64	1024	this_job:reachable	3.50e-04	0.0374	0.0308	yes	3780165
32	rhow64	1208	this_job:reachable	2.40e-04	0.0325	0.0280	yes	3780165
32	std	256	this_job:reachable	8.60e-02	0.6639	0.6101	no	3780165
32	std	512	this_job:reachable	4.14e-03	0.2339	0.1944	no	3780165
32	std	1024	this_job:reachable	3.40e-04	0.0481	0.0329	yes	3780165
32	std	2048	this_job:reachable	3.23e-05	0.0073	0.0049	yes	3780165
32	std	2628	this_job:reachable	1.23e-05	0.0058	0.0029	yes	3780165
32	std	2675	this_job:reachable	1.24e-05	0.0062	0.0021	yes	3780165

Table F.23: Every quadrature rule at $q \geq 64$ (continuation of Table F.22; same columns and bars).

q	fit arm	m	population	NNLS rel. fit	$\rho_{\max}$	ρ_{95}	primary	job
64	reachable	1024	qrg304:reachable	7.12e-04	0.0531	0.0230	yes	3780164
64	reachable	1024	qrg304:reachable	7.12e-04	0.0531	0.0230	yes	3780165
64	reachable	1024	qrg304:reachable	7.12e-04	0.0531	0.0230	yes	3783811
64	reachable	2048	qrg304:reachable	7.54e-05	0.1120	0.0172	yes	3780164
64	reachable	2048	qrg304:reachable	7.54e-05	0.1120	0.0172	yes	3780165
64	reachable	2048	qrg304:reachable	7.54e-05	0.1120	0.0172	yes	3783811
64	static	1280	qrg304:static	2.43e-04	0.1439	0.0235	no	3780164
64	static	1280	qrg304:static	2.43e-04	0.1439	0.0235	no	3780165
64	static	1280	qrg304:static	2.43e-04	0.1439	0.0235	no	3783811
64	fs64	1024	this_job:reachable	9.19e-04	0.1303	0.0332	no	3780165
64	fs64	2048	this_job:reachable	1.14e-04	0.1248	0.0032	no	3780165
64	repro64m2048s1	2048	this_job:reachable	1.26e-04	0.0459	0.0088	yes	3783811
64	repro64m2048s2	2048	this_job:reachable	1.18e-04	0.1036	0.0090	yes	3783811
64	repro64m2048s3	2048	this_job:reachable	1.27e-04	0.0134	0.0042	yes	3783811
64	repro64m2048s4	2048	this_job:reachable	1.36e-04	0.0108	0.0052	yes	3783811
64	reproincumbentm1024s1	1024	this_job:reachable	4.85e-04	0.1420	0.0608	no	3783811
64	reproincumbentm1024s2	1024	this_job:reachable	4.66e-04	0.1509	0.0526	no	3783811
64	reproincumbentm1024s3	1024	this_job:reachable	6.86e-04	0.1587	0.0924	no	3783811
64	reproincumbentm1024s4	1024	this_job:reachable	6.26e-04	0.0670	0.0309	yes	3783811
64	rhov64	1024	this_job:reachable	8.60e-04	0.1039	0.0364	yes	3780165
64	rhov64	1563	this_job:reachable	2.48e-04	0.0518	0.0238	yes	3780165
64	std	256	this_job:reachable	1.84e-01	0.7670	0.7632	no	3780165
64	std	512	this_job:reachable	1.28e-02	0.3106	0.2780	no	3780165
64	std	1024	this_job:reachable	6.97e-04	0.1220	0.0262	no	3780165
64	std	2048	this_job:reachable	6.75e-05	0.0583	0.0074	yes	3780165
64	std	3054	this_job:reachable	1.18e-05	0.0213	0.0022	yes	3780165
64	std	3128	this_job:reachable	1.11e-05	0.0129	0.0023	yes	3780165
128	reachable	1024	qrg304:reachable	1.62e-03	0.2786	0.2246	no	3780164
128	reachable	1024	qrg304:reachable	1.62e-03	0.2786	0.2246	no	3780165
128	reachable	1024	qrg304:reachable	1.62e-03	0.2786	0.2246	no	3783811
128	reachable	2048	qrg304:reachable	1.04e-04	0.1908	0.0518	no	3780164
128	reachable	2048	qrg304:reachable	1.04e-04	0.1908	0.0518	no	3780165
128	reachable	2048	qrg304:reachable	1.04e-04	0.1908	0.0518	no	3783811
128	static	2048	qrg304:static	7.16e-05	0.2194	0.0677	no	3780164
128	static	2048	qrg304:static	7.16e-05	0.2194	0.0677	no	3780165
128	static	2048	qrg304:static	7.16e-05	0.2194	0.0677	no	3783811
128	fs64	2048	this_job:reachable	3.52e-04	0.0669	0.0214	yes	3780164
128	fs64	2560	this_job:reachable	1.82e-04	0.0292	0.0111	yes	3780164
128	repro64m2048s1	2048	this_job:reachable	4.12e-04	0.0497	0.0372	yes	3783811
128	repro64m2048s2	2048	this_job:reachable	4.31e-04	0.0426	0.0298	yes	3783811
128	repro64m2048s3	2048	this_job:reachable	3.25e-04	0.0788	0.0234	yes	3783811
128	repro64m2048s4	2048	this_job:reachable	3.66e-04	0.2040	0.0506	no	3783811
128	reproincumbentm1024s1	1024	this_job:reachable	1.62e-03	0.3523	0.3057	no	3783811
128	reproincumbentm1024s2	1024	this_job:reachable	1.70e-03	0.2098	0.1380	no	3783811
128	reproincumbentm1024s3	1024	this_job:reachable	1.01e-03	0.3357	0.2786	no	3783811
128	reproincumbentm1024s4	1024	this_job:reachable	1.36e-03	0.4609	0.4217	no	3783811
128	rhov64	2048	this_job:reachable	3.54e-04	0.0472	0.0176	yes	3780164
128	rhov64	2319	this_job:reachable	2.52e-04	0.0277	0.0102	yes	3780164
128	std	1024	this_job:reachable	1.49e-03	0.2793	0.1615	no	3780164
128	std	2048	this_job:reachable	7.98e-05	0.2373	0.0520	no	3780164
128	std	2560	this_job:reachable	3.14e-05	0.2040	0.0308	no	3780164
128	std	3055	this_job:reachable	1.40e-05	0.1193	0.0197	no	3780164
128	std	3088	this_job:reachable	1.35e-05	0.1091	0.0183	yes	3780164
128	std	3128	this_job:reachable	1.33e-05	0.1487	0.0161	no	3780164
256	reachable	1024	qrg304:reachable	4.01e-02	0.5731	0.5370	no	3780164
256	reachable	1024	qrg304:reachable	4.01e-02	0.5731	0.5370	no	3780165
256	reachable	1024	qrg304:reachable	4.01e-02	0.5731	0.5370	no	3783811
256	reachable	2048	qrg304:reachable	3.08e-04	0.1678	0.0957	no	3780164
256	reachable	2048	qrg304:reachable	3.08e-04	0.1678	0.0957	no	3780165
256	reachable	2048	qrg304:reachable	3.08e-04	0.1678	0.0957	no	3783811
256	static	2048	qrg304:static	1.49e-04	0.4621	0.4275	no	3780164
256	static	2048	qrg304:static	1.49e-04	0.4621	0.4275	no	3780165
256	static	2048	qrg304:static	1.49e-04	0.4621	0.4275	no	3783811
256	fs64	2048	this_job:reachable	2.41e-03	0.1074	0.0878	yes	3780164
256	fs64	2560	this_job:reachable	9.35e-04	0.0647	0.0314	yes	3780164
256	repro64m2048s1	2048	this_job:reachable	1.34e-03	0.1820	0.1299	no	3783811
256	repro64m2048s2	2048	this_job:reachable	1.64e-03	0.1899	0.1209	no	3783811
256	repro64m2048s3	2048	this_job:reachable	1.41e-03	0.1299	0.0909	no	3783811
256	repro64m2048s4	2048	this_job:reachable	1.33e-03	0.2421	0.1413	no	3783811
256	reproincumbentm1024s1	1024	this_job:reachable	1.21e-02	0.6743	0.4499	no	3783811
256	reproincumbentm1024s2	1024	this_job:reachable	2.26e-02	0.5737	0.5502	no	3783811
256	reproincumbentm1024s3	1024	this_job:reachable	3.35e-02	0.7319	0.7216	no	3783811
256	reproincumbentm1024s4	1024	this_job:reachable	1.71e-02	0.5075	0.4916	no	3783811
256	rhov64	2048	this_job:reachable	1.44e-03	0.1299	0.0836	no	3780164
256	rhov64	2560	this_job:reachable	6.23e-04	0.1018	0.0380	yes	3780164
256	rhov64	3072	this_job:reachable	3.37e-04	0.0963	0.0229	yes	3780164
256	std	1024	this_job:reachable	1.61e-02	0.4977	0.4450	no	3780164
256	std	2048	this_job:reachable	1.75e-04	0.2958	0.1822	no	3780164
256	std	2560	this_job:reachable	5.84e-05	0.1956	0.1052	no	3780164
256	std	3072	this_job:reachable	2.32e-05	0.1774	0.0722	no	3780164
256	std	3370	this_job:reachable	1.45e-05	0.1308	0.0685	no	3780164
256	std	3410	this_job:reachable	1.43e-05	0.1761	0.0836	no	3780164

Table F.24: The EQ ladder with the cheapest rule passing the primary bar in its draw per rung, timed in one allocation (job 3780164, A100 80 GB), with its same-job dense twins, and the construction status from the four-draw replication (job 3783811): confirmed = every re-draw passes, marginal = only the listed fraction does. Errors are as measured with these rules (SHA256-identified); they are not properties of the construction above $q = 32$. Ladder rows read from the lane summary.json (final).

q	m	fit states	$\rho_{\max}$ (this draw)	construction status	EQ evolved %	EQ all %	EQ ms	dense evolved %	dense ms	EQ/dense cost
0	1024	64	0.0153	confirmed (3 of 3 re-draws)	1.8891	2.5629	59.1	1.8890	292.9	0.202
16	1024	64	0.0935	confirmed (3 of 3 re-draws)	1.4270	2.4806	80.6	—	—	—
32	1024	42	0.0533	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.7	—	—	—
64	1024	25	0.0531	marginal (2 of 6 draws pass)	1.2275	2.1489	120.6	1.0843	624.9	0.193
128	2048	64	0.0669	marginal (4 of 5 draws pass)	0.8936	1.8116	246.9	0.8930	1190.5	0.207
256	2048	64	0.1074	marginal (1 of 5 draws pass)	0.5389	0.9053	722.2	0.5194	3915.1	0.184

Table F.25: The pre-registered draw replication (job 3783811): four independent draws of candidate pool and fit-state subset at fixed q , m , fit-state count and scaling, with the number of draws passing each bar and the construction status over every draw of that construction in all three jobs. Withdrawn by these data: the parent lane’s extrapolated crossing node count; the inherited fit-state convention for $q \geq 128$ (0 of 6 draws pass); “64 states is uniformly better” (1 of 5 at $q = 256$); and the interim rule export at $q \geq 64$, superseded by rules chosen by moving up in m , a post-hoc choice the lane discloses.

q	m	fit states	four draws: $\rho_{\max}$	min / median / max	spread	primary	tight	construction	status (all draws)
64	1024	25	0.1420, 0.1509, 0.1587, 0.0670	0.0670 / 0.1464 / 0.1587	2.37	1/4	0/4	marginal (2 of 6 draws pass)	
64	2048	64	0.0459, 0.1036, 0.0134, 0.0108	0.0108 / 0.0297 / 0.1036	9.57	4/4	3/4	marginal (4 of 5 draws pass)	
128	1024	14	0.3523, 0.2098, 0.3357, 0.4609	0.2098 / 0.3440 / 0.4609	2.20	0/4	0/4	marginal (0 of 6 draws pass)	
128	2048	64	0.0497, 0.0426, 0.0788, 0.2040	0.0426 / 0.0642 / 0.2040	4.79	3/4	2/4	marginal (4 of 5 draws pass)	
256	1024	8	0.6743, 0.5737, 0.7319, 0.5075	0.5075 / 0.6240 / 0.7319	1.44	0/4	0/4	marginal (0 of 6 draws pass)	
256	2048	64	0.1820, 0.1899, 0.1299, 0.2421	0.1299 / 0.1860 / 0.2421	1.86	0/4	0/4	marginal (1 of 5 draws pass)	

Table F.26: Frozen-checkpoint mesh ladder (Burgers job 3711388on NVIDIA A100-PCIE-40GB, checkpoint 18f0266ae6f0...; Poisson job 3711389on NVIDIA A100 80GB PCIe, checkpoint a128e7635c31...). “efficient FOM” is the cheapest same-job full-order arm meeting the 5 % target on that mesh.

PDE	intervals	unknowns	ROM cached ms	ROM device ms	ROM complete-query ms	ROM worst vs ref %	ROM vs same-grid FOM %	efficient FOM	FOM ms	FOM/ROM	ROM meets 5%
Burgers	64	3,969	44.2	48.5	50.4	10.8570	5.5018	fom_same_ntile-2	12.9	0.265	no
Burgers	128	16,129	44.7	46.5	48.1	6.5431	2.8939	fom_same_ntile-2	14.9	0.320	no
Burgers	256	65,025	44.7	48.2	50.6	4.5546	2.5629	fom_same_ntile-2	15.7	0.325	yes
Burgers	512	261,121	45.1	47.4	53.7	3.7168	3.2836	fom_same_ntile-2	23.5	0.495	yes
Burgers	1024	1,046,529	43.6	50.1	76.8	3.8847	3.8362	fom_coarsest_ntile-4	22.2	0.443	yes
Poisson	64	3,969	2.1	2.3	3.2	6.1119	6.1282	dst	0.1	0.063	no
Poisson	128	16,129	2.1	2.2	3.2	6.1107	6.1149	dst	0.1	0.065	no
Poisson	256	65,025	2.1	2.2	3.3	6.1106	6.1116	dst	0.2	0.067	no
Poisson	512	261,121	1.9	2.3	3.8	6.1106	6.1108	dst	0.2	0.079	no
Poisson	1024	1,046,529	2.0	3.0	6.6	6.1106	6.1106	dst	0.3	0.112	no

Table F.27: Reflective 2D wave, one job per mesh (64^2 : job 3780447 (NVIDIA A100 80GB PCIe, commit 0bb3cc86); 256^2 : job 3783805 (NVIDIA A100-PCI-E-40GB, commit 2655bb01); 1024^2 : job 3780450 (NVIDIA A100 80GB PCIe, commit 0bb3cc86)); costs comparable only within a mesh. The top rung is the learned bank evolved by exact modal propagation with no head.

mesh	arm	q / k'	worst energy-state %	$t=0$ %	device ms	complete-query ms
64^2	head_q0	0	11.2055	5.5314	184.535	185.575
64^2	trained_nested40	8	6.8128	4.7478	200.232	201.072
64^2	nested_q8	8	5.9379	3.7715	199.289	200.375
64^2	nested_q16	16	5.0053	2.5994	1977.402	1978.383
64^2	nested_q32	32	4.9161	2.0345	2500.060	2501.038
64^2	linear_bank64	$64 (= R)$	4.9323	2.0345	1.168	1.986
64^2	pod_k40	$k'=40$	4.8825	4.2685	1.190	1.989
64^2	pod_k64	$k'=64$	1.4893	1.4709	1.202	2.018
64^2	pod_k128	$k'=128$	0.8668	0.4932	1.252	2.052
64^2	dst	—	0.0000	0.0000	3.391	4.286
64^2	rk4_fom	—	0.0612	0.0000	5.511	6.865
64^2	cg_1e-06	—	0.5745	0.0000	109.836	111.350
256^2	head_q0	0	11.3314	5.5585	183.516	198.619
256^2	trained_nested40	8	6.8764	4.7840	197.587	213.401
256^2	nested_q8	8	6.0627	3.8045	197.389	212.596
256^2	nested_q16	16	5.1661	2.6270	1989.463	2004.700
256^2	nested_q32	32	5.1038	2.0731	2520.177	2536.035
256^2	linear_bank64	$64 (= R)$	5.1208	2.0731	1.428	16.659
256^2	pod_k40	$k'=40$	4.9963	4.2844	1.365	16.639
256^2	pod_k64	$k'=64$	1.5018	1.4817	1.399	16.657
256^2	pod_k128	$k'=128$	0.9873	0.5122	1.482	16.773
256^2	dst	—	0.0000	0.0000	4.356	19.703
256^2	rk4_fom	—	0.0016	0.0000	24.346	40.083
256^2	cg_1e-06	—	0.6018	0.0000	142.511	158.115
1024^2	head_q0	0	11.3388	5.5602	192.462	451.994
1024^2	trained_nested40	8	6.8789	4.7863	207.832	467.228
1024^2	nested_q8	8	6.0692	3.8066	206.632	468.808
1024^2	nested_q16	16	5.1763	2.6287	2003.425	2264.489
1024^2	nested_q32	32	5.1164	2.0755	2529.445	2789.801
1024^2	linear_bank64	$64 (= R)$	5.1335	2.0755	4.562	265.785
1024^2	pod_k40	$k'=40$	5.0030	4.2854	4.183	263.883
1024^2	pod_k64	$k'=64$	1.5010	1.4824	4.232	264.860
1024^2	pod_k128	$k'=128$	0.9888	0.5135	5.879	265.409
1024^2	dst	—	0.0000	0.0000	17.103	277.324
1024^2	rk4_fom	—	0.0000	0.0000	958.007	1174.433
1024^2	cg_1e-06	—	0.6079	0.0000	1043.368	1278.837

Table F.28: Poisson head capacity on the frozen $R = 512$ bank (job 3783883, NVIDIA A100-PCI-E-40GB): width helps, depth alone does not, and no arm reaches a $2\times$ ratio to the floor. The re-run of the primary recipe reproduces it exactly.

head	best-found dev. %	best-found / floor	solved 1024^2 %	complete-query ms
K32_w128_L2	3.1212	4.18	3.1146	12.93
K32_w256_L2	2.2652	3.04	2.2600	12.84
K32_w128_L3	3.0625	4.11	3.0558	13.23
K32_w256_L3	2.3614	3.17	2.3571	13.39
K64_w128_L2	2.5625	3.44	2.5587	13.72
K64_w256_L3	2.0553	2.76	2.0549	14.12
K32_w128_L2_x3	2.6428	3.54	2.6360	12.97

Table F.29: Three training seeds on the development cohort (lane b-seeds, jobs `seed1` = 3783776 (NVIDIA A100 80GB PCIe); `seed2` = 3783777 (NVIDIA A100 80GB PCIe); `seed3` = 3783778 (NVIDIA A100-PCIE-40GB)). Top: per rung of the dense $M = 4(K + q)$ ladder, worst error over the six development cases as mean $\pm$ sample standard deviation over the 3 seed checkpoints, beside the incumbent checkpoint re-run in each seed's own job (its value is identical in all three jobs). Bottom: the per-seed ladder verdicts. Costs are never averaged across jobs. This is the development cohort; the sealed cohort (Table F.30) is the headline.

q	M	evolved %, seeds	evolved %, incumbent	all-times %, seeds	all-times %, incumbent	best-found %, seeds	converged
0	64	2.0002 $\pm$ 0.4986	1.8890	2.6659 $\pm$ 0.1715	2.5629	2.6552 $\pm$ 0.1781	3 of 3
16	128	1.5988 $\pm$ 0.0995	1.3985	2.5842 $\pm$ 0.1534	2.4806	2.5731 $\pm$ 0.1576	3 of 3
32	192	1.3727 $\pm$ 0.1314	1.2336	2.4971 $\pm$ 0.1344	2.3534	2.4928 $\pm$ 0.1359	3 of 3
64	320	1.2130 $\pm$ 0.0580	1.0843	2.2759 $\pm$ 0.0790	2.1489	2.2632 $\pm$ 0.0946	3 of 3
128	576	0.9562 $\pm$ 0.0264	0.8930	1.8806 $\pm$ 0.0704	1.8116	1.8793 $\pm$ 0.0704	3 of 3
256	1088	0.4987 $\pm$ 0.1164	0.5194	0.9538 $\pm$ 0.0381	0.9053	0.9431 $\pm$ 0.0398	2 of 3

seed	job	GPU	monotone (evolved)	monotone (all-times)	every rung converged	error span	cost span	knob bar
seed1	3783776	NVIDIA A100 80GB PCIe	yes	yes	yes	4.07 $\times$	16.26 $\times$	yes
seed2	3783777	NVIDIA A100 80GB PCIe	yes	yes	yes	4.01 $\times$	10.34 $\times$	yes
seed3	3783778	NVIDIA A100-PCIE-40GB	yes	yes	no	3.93 $\times$	20.97 $\times$	no

Table F.30: The sealed cohort (lane b-seeds, job 3804465, NVIDIA A100-PCIE-40GB), opened once after every choice was frozen. Top: per rung of the dense $M = 4(K + q)$ ladder, the incumbent's sealed worst evolved error and device time, the seed mean $\pm$ sample standard deviation on the sealed and the development cohort, the lane's sealed-over-development ratio of seed means (raw and difficulty-normalised) and the converged count over the 4 checkpoints. Bottom: per-checkpoint verdicts. The incumbent's $q = 0$ rung is a single sealed case converged to a wrong branch (gradient exit, zero budget exits); the pre-registered ratio criterion fails at $q = 0$ for the incumbent alone (5.35) and the convergence criterion fails on `seed2` at $q = 64$ (3 budget exits).

q	M	incumbent sealed %	ms	seeds sealed %	seeds development %	sealed / dev	normalised	converged
0	64	10.1120	375.1	2.8023 $\pm$ 0.6357	2.0002 $\pm$ 0.4986	1.40	1.41	4 of 4
16	128	1.4617	480.8	1.7305 $\pm$ 0.1297	1.5988 $\pm$ 0.0995	1.08	1.09	4 of 4
32	192	1.4458	540.5	1.5044 $\pm$ 0.1422	1.3727 $\pm$ 0.1314	1.10	1.14	4 of 4
64	320	1.3252	754.5	1.3528 $\pm$ 0.1099	1.2130 $\pm$ 0.0580	1.12	1.13	3 of 4
128	576	1.0964	1460.4	1.0461 $\pm$ 0.0619	0.9562 $\pm$ 0.0264	1.09	1.17	4 of 4
256	1088	0.6789	6725.5	0.6188 $\pm$ 0.0345	0.4987 $\pm$ 0.1164	1.24	1.20	4 of 4

checkpoint	job	monotone (evolved)	monotone (all-times)	every rung converged	error span	cost span	knob bar
incumbent	3804465	yes	yes	yes	14.89 $\times$	17.93 $\times$	yes
seed1	3804465	yes	yes	yes	3.52 $\times$	13.25 $\times$	yes
seed2	3804465	yes	yes	no	4.81 $\times$	13.17 $\times$	no
seed3	3804465	yes	yes	yes	5.22 $\times$	14.62 $\times$	yes

Table F.31: The operators' own inference-time knob: each validation-selected checkpoint evaluated at coarser grids and prolonged back (job 3787189, NVIDIA A100 80GB PCIe), with the interpolation floor (a perfect operator's error at that rung) and the pre-registered gates ($\geq 1.5\times$ own speed at $\leq 2\times$ own error). Speedups are within one model's own curve, same job.

operator	rung	worst evolved %	interp. floor %	device ms	speed vs 256	error vs 256	both gates
fno-large	256	6.38	0.00	7.35	—	—	ref.
fno-large	128	7.30	0.23	3.46	2.12	1.14	yes
fno-large	64	9.82	0.92	3.44	2.14	1.54	yes
fno-large	32	13.85	4.02	3.33	2.21	2.17	no
tsol-refine	256	9.32	0.00	11.20	—	—	ref.
tsol-refine	128	15.29	0.23	8.15	1.37	1.64	no
tsol-refine	64	35.18	0.92	8.09	1.38	3.78	no
tsol-refine	32	75.13	4.02	8.04	1.39	8.06	no
unet-refine	256	7.52	0.00	5.90	—	—	ref.
unet-refine	128	38.13	0.23	2.35	2.51	5.07	no
unet-refine	64	60.18	0.92	2.31	2.55	8.01	no
unet-refine	32	67.65	4.02	2.30	2.56	9.00	no

Table F.32: One-variable controls on the validation-selected capacities (job 3783831): each repeats a screen arm under the identical protocol and budget with exactly one variable changed, and is never eligible for capacity selection. "vs ROM" is the matched eight-case worst against the ROM's, from job 3702709; at equal wall clock a float64 twin is also a fewer-epochs twin.

control	twin	one variable changed	epochs vs twin	control worst %	twin worst %	control vs ROM	twin vs ROM
ctrl-medium-f64	unet-medium	network dtype float64	500 vs 1963 (0.25 $\times$)	1.6721	1.5189	below (0.1950 pp)	below
ctrl-medium-seed2	unet-medium	training seed	1969 vs 1963 (1.00 $\times$)	1.5479	1.5189	below (0.3192 pp)	below
ctrl-tsol-small-f64	tsol-small	network dtype float64	682 vs 1628 (0.42 $\times$)	3.1329	2.5273	above (1.2658 pp)	above

Table F.33: U-Net against FNO on the Poisson operator-screen dataset (physical reference). The U-Net arms hit the epoch cap while still improving; the FNO early-stopped.

arm	family	params	median %	worst %	job
fno-large	FNO	17,876,673	2.03	29.37	3702464
fno-medium	FNO	5,779,729	2.35	28.28	3702464
fno-small	FNO	1,192,801	2.64	32.47	3702464
unet-large	U-Net	17,461,393	1.95	8.31	3780625
unet-medium	U-Net	7,763,041	1.68	6.02	3780625
unet-small	U-Net	4,368,073	2.40	12.80	3780625

Table F.34: Speed at bit-level parity (jobs 3745655 (spd01), 3745656 (fine01), 3745913 (comp01)): fastest arm whose fields agree with the incumbent to 10^{-12} with identical iteration counts and exit reasons. The pre-registered $2\times$ target fails at every mesh.

attempt	intervals	arm	incumbent ms	arm ms	GPU speedup	incl. host	parity	$2\times$ target
spd01	256	L4	46.856	30.824	1.520x	1.478x	2.565e-13	FAIL
fine01	512	L4	46.928	31.106	1.509x	1.454x	5.053e-13	FAIL
fine01	1024	L4	49.283	33.544	1.469x	1.310x	6.951e-13	FAIL
comp01	256	C1	46.627	30.723	1.518x	1.482x	2.562e-13	FAIL
comp01	1024	C1	49.804	33.282	1.496x	1.408x	7.129e-13	FAIL

Table F.35: Training study on the Burgers head (jobs 3745912 (training), 3749074 (evaluation)): data density, objective, latent dimension and a joint bank arm, each evaluated through the unchanged head-ablation machinery. 0 of 26 arms pass the pre-registered success criterion. The like-for-like retrain does not reproduce the incumbent, so the density verdict is measured below baseline.

arm (EQ query)	K	bank floor %	best-found %	solved all %	solved evolved %	device ms	conv.
incumbent (4608 traj., $K=16$)	16	0.3918	2.5447	2.5629	1.9002	48.7	yes
128 traj., $K=16$	16	0.3918	12.6496	12.6496	7.2529	45.4	yes
512 traj.	16	0.3918	6.8202	6.8204	3.7153	39.3	yes
2048 traj.	16	0.3918	3.5570	3.5574	1.8752	46.6	yes
4608 traj. (like-for-like retrain)	16	0.3918	3.9616	3.9710	1.9522	59.1	yes
128 traj., $K=32$	32	0.3918	8.8489	8.8496	6.2351	80.7	yes
2048 traj., $K=32$	32	0.3918	3.1132	3.1137	2.4882	54.5	yes
128 traj., weak-residual term	16	0.3918	14.2164	14.2165	7.5935	43.9	yes
128 traj., trajectory term	16	0.3918	12.8237	12.8238	7.8543	42.8	yes
128 traj., code-smoothness term	16	0.3918	13.6497	13.6505	13.4135	36.8	yes
selected: 2048 traj., $K=32$, weak term	32	0.3918	2.8289	3.1275	3.1275	55.1	yes
joint bank+head, $R=512$	32	2.8517	5.0383	5.0471	2.4399	46.3	yes

Table F.36: Navier–Stokes 2D, phase-2 gates in four settings: the $K = 16$ head on the full-rank $R = 256$ bank, the $K = 32$ head on the full-rank $R = 512$ bank, the same $K = 16$ recipe on a family of intrinsic dimension 8 instead of 14 (3 modes; job 3808498, A100 (pax105)), and the $K = 16$ recipe at $4\times$ the training data (2048 trajectories: 512 gated plus 1536 extra, same family; last row, job 3808495, A100 (pax106)) (jobs 3787319 ($K=16$, $R=256$), 3787320 ($K=32$, $R=512$), 3808498 ($K=16$, $R=256$, family dimension 8), 3808495 ($K=16$, $R=256$, 2048 trajectories); full-order solver and dataset certified in job 3780151). Every gate passes except the held-out oracle, whose pre-registered bar was a $2.0\times$ margin over POD at matched dimension (ns2d/DESIGN.md H-ORACLE gate: oracle median $\leq 1/2 \times$ POD-K held-out, i.e. ratio ≥ 2.0 ; pre-registered at commit 1281ff73 (2026-09-17 00:55), before jobs 3783796/3787319/3787320); the ratio is 1.15–1.46 across the 4 settings, so the ladder and timing were never run for any head and the lane closed without a phase-3 job. With $4\times$ the data the training reconstruction rises from 5.0345 to 12.3875 % and the held-out/training gap falls from 4.0 to 1.3: the head no longer fits the training data either, so the limit moved from generalisation to capacity at the same head (0.52 at $t = 0$, 1.39 on evolved times). The $4\times$ -data arm’s B-DATA gate passed by the pre-registered value path (a hash mismatch on its node, at 10^{-16} on the archived trajectories). On the lower-dimensional family every error falls $3.3\text{--}4.4\times$ (POD-16 3.3, oracle 3.8, bank floor 4.4) while the ratio stays at 1.39 (1.83 at $t = 0$, 1.44 on evolved times) and the held-out/training gap at 3.8 against 4.0. Oracle values are upper bounds: 12 ($K=16$), 18 ($K=32$), 4 (family dimension 8) and 6 (2048 trajectories) held-out fits at 256^2 hit the LM budget, which biases the ratio downward, so the negative verdict is conservative.

mesh	bank rank	B-ORTH	bank worst %	POD- R worst %	B-FLOOR	oracle median %	POD- K median %	POD- K / oracle	H-ORACLE (≥ 2.0)
64^2	256	yes	14.0487	13.0669	yes	20.3442	24.1747	1.19	no
128^2	256	yes	14.0932	13.0156	yes	20.2574	24.0400	1.19	no
256^2	256	yes	14.0838	13.0322	yes	20.2417	24.0219	1.19	no
64^2 ($K=32$)	512	yes	7.4103	6.0761	yes	12.1757	14.0697	1.16	no
128^2 ($K=32$)	512	yes	7.5545	5.9861	yes	12.1115	13.9568	1.15	no
256^2 ($K=32$)	512	yes	7.5643	5.9983	yes	12.0948	13.9455	1.15	no
256^2 (family dim. 8)	256	yes	6.4018	6.2295	yes	5.2823	7.3651	1.39	no
256^2 (2048 traj.)	256	yes	13.8371	11.5733	yes	16.6598	24.2492	1.46	no

Table F.37: L-shaped Poisson, solve layer at $M = 257$ (jobs 3784662, 3784663, 3789568, one per mesh): the non-dominated set on (complete-query ms, worst same-grid error) at each mesh, over every reduced and full-order subject. The sparse direct solve is exact to round-off. The 512^2 job's first attempt (Table E.2) failed a residual gate that is unreachable in float64 at that mesh, where the achievable residual floor grows like N^2 ; the gate now takes the larger of the two bounds, which changed no mesh already run, and the re-run differs from that attempt only by the gate.

mesh	subject	family	worst same-grid %	complete-query ms
64^2	sparse direct (SuperLU)	fom	0.0000	1.282
128^2	POD $k'=16$	pod	34.6279	2.086
128^2	POD $k'=32$	pod	19.0053	2.233
128^2	POD $k'=64$	pod	7.7013	2.294
128^2	sparse direct (SuperLU)	fom	0.0000	2.475
256^2	POD $k'=16$	pod	34.5764	2.240
256^2	POD $k'=32$	pod	18.9497	2.341
256^2	POD $k'=64$	pod	7.6645	2.483
256^2	head $q=0$ (head_smooth_R256_K16)	neural	3.4928	2.848
256^2	POD $k'=128$	pod	2.4575	2.850
256^2	head $q=64$ (head_sdf_R512_K16)	neural+linear	2.1305	3.028
256^2	sparse direct (SuperLU)	fom	0.0000	8.386
512^2	POD $k'=16$	pod	34.5639	3.217
512^2	POD $k'=32$	pod	18.9358	3.390
512^2	POD $k'=64$	pod	7.6554	3.512
512^2	POD $k'=128$	pod	2.4511	4.031
512^2	head $q=128$ (head_sdf_R512_K16)	neural+linear	2.1977	4.748
512^2	head $q=64$ (head_sdf_R512_K16)	neural+linear	2.1233	4.801
512^2	CG 10^{-2}	fom	0.3845	29.127
512^2	sparse direct (SuperLU)	fom	0.0000	36.505

Table F.38: L-shaped Poisson, free rung at $M = 1024$ (job 3784910). **These costs are not comparable with Table F.37:** the dense projection is charged inside every reduced query here. Non-dominated set only.

subject	family	worst same-grid %	complete-query ms
POD $k'=16$	pod	34.5764	2.451
POD $k'=32$	pod	18.9497	2.458
free rung $q=R$ (head_sdf_R512_K32)	free	0.7791	2.582
free rung $q=R$ (head_sdf_R512_K16)	free	0.7791	2.583
sparse direct (SuperLU)	fom	0.0000	8.351

Table F.39: L-shaped Poisson (job 3783786, NVIDIA A100 80GB PCIe): bank floors by boundary factor and rank, and the head layer at 256^2 . The solve layer is in Table F.37 (64^2 – 512^2 , one job per mesh).

bank	floor, dev. 256^2 %	floor, dev. 512^2 %	floor, common 256^2 %
smooth_R256	1.5113	1.5057	3.1933
smooth_R512	0.7123	0.7093	1.9344
sdf_R256	1.5697	1.5626	3.2101
sdf_R512	0.7766	0.7698	1.6155
enrich_R512	0.6676	0.6650	1.7028
smooth_ff128s2_R512	2.5438	2.5388	3.8411

head arm	best-found %	weak solve % (untimed)	best-found / bank floor
head_enrich_R512_K16	3.8508	3.8510	5.77
head_sdf_R256_K16	3.5177	3.5181	2.24
head_sdf_R512_K16	3.8607	3.8608	4.97
head_sdf_R512_K32	3.1802	3.1814	4.09
head_smooth_R256_K16	3.4919	3.4928	2.31
head_smooth_R512_K16	3.5924	3.5931	5.04
head_smooth_ff128s2_R512_K16	6.7961	6.7970	2.67

Table G.1: Navier–Stokes head-only data scaling (job 3808493): held-out oracle median per recipe and training-set size, POD-16 at matched dimension, the held-out/train ratio, the selected training step, and the pre-registered log-log slope with its verdict (band -0.25 to -0.10).

recipe	training trajectories	held-out oracle median %	POD-16 median %	POD / oracle	held-out / train	selected step
plain recipe	128	27.4938	23.5757	0.86	16.5	100000
plain recipe	256	20.6889	22.9610	1.11	6.9	100000
plain recipe	512	19.8279	22.7726	1.15	4.1	100000
plain recipe (all n)	—	slope -0.236	—	—	—	ambiguous
weight decay + early stopping	128	26.1095	23.5757	0.90	4.6	5000
weight decay + early stopping	256	21.0204	22.9610	1.09	3.2	10000
weight decay + early stopping	512	19.3892	22.7726	1.17	3.1	30000
weight decay + early stopping (all n)	—	slope -0.215	—	—	—	ambiguous
weight decay, head $4 \times$ smaller	128	24.6583	23.5757	0.96	3.4	95000
weight decay, head $4 \times$ smaller	256	21.3431	22.9610	1.08	1.8	10000
weight decay, head $4 \times$ smaller	512	20.0874	22.7726	1.13	1.6	45000
weight decay, head $4 \times$ smaller (all n)	—	slope -0.148	—	—	—	ambiguous

Table G.2: Navier–Stokes, **exploratory after a failed phase-2 gate** (job 3808502): the correction ladder on the $K = 32$, $R = 512$ manifold at fixed $M = 2176$, its three-layer decomposition (manifold layer and bank floor as the median over cases of the worst evolved time, and the solved/manifold ratio on that matched statistic), the share of head-residual energy the q directions capture, and same-job POD-LSPG at the matched dimension $k' = K + q$; costs are median ms inside this job. “Non-dom.” marks membership of the non-dominated set on (ms, worst evolved error) over every subject in the job.

q	$k' = K + q$	neural worst %	median %	ms	budget exits	manifold layer %	bank floor %	solved / manifold	energy %	POD- k' worst %	median %	ms	neural non-dom.	POD non-dom.
0	32	68.7654	41.3922	13387.5	0	16.4787	1.7434	2.51	0	54.6331	33.4692	212.1	—	yes
32	64	70.7701	36.8058	16979.1	0	15.9678	1.7434	2.31	26	35.5092	19.0979	431.0	—	—
64	96	61.5127	32.7278	21289.9	0	14.8960	1.7434	2.20	44	27.7285	13.2380	832.7	—	—
128	160	56.5456	25.6655	29011.5	0	13.4634	1.7434	1.91	68	17.1958	8.0974	2398.2	—	—
256	288	33.0536	13.8903	46267.9	0	6.8226	1.7434	2.04	92	9.5434	3.9876	9963.9	—	—
512	544	6.0140	2.4577	52592.3	0	1.7434	1.7434	1.40	100	4.1664	1.4871	53902.5	—	—

Table G.3: Navier–Stokes, **exploratory after a failed phase-2 gate** (job 3808502): the full-order Newton-tolerance ladder timed in the same job as Table G.2; the last row is the converged reference.

Newton tol.	worst evolved %	median %	ms	non-dominated
0.03	106.4413	95.1088	46.7	yes
0.01	106.4413	95.1088	47.2	—
0.003	25.5895	7.3378	263.2	yes
0.001	0.0041	0.0030	421.3	yes
0.0001	0.0033	0.0006	425.6	yes
1e-06	0.0033	0.0007	477.8	yes
1e-11	0.0000	0.0000	1177.7	yes

Table G.4: Secondary comparison against the previous submission’s comparator, unpreconditioned CG, beside the competitive one, all inside one job per mesh: the reduced rungs (and the linear top rung and POD) with their complete-query ms, the CG solve at the tolerance shown, the direct solve (DST on the square, SuperLU on the L-shape) and the ratios. Poisson from the p-linear jobs, L-shape from the lshape jobs (CG at its own tolerances, the tightest shown; the IC(0)-PCG arm runs on the CPU). Burgers has no CG comparator (its full-order solver is Newton). Heat is omitted: its committed CG record is per-invocation raw timing of an earlier decoder family and is not re-aggregated here.

mesh	subject	error %	ms	CG tol.	CG ms	CG / ROM	direct	direct ms	direct / ROM	IC(0)-PCG (CPU) ms	PCG / ROM
256 ² (job 3780692)	$q=0$	3.1567	9.214	10^{-6}	38.05	4.1	DST	2.526	0.27	—	—
256 ² (job 3780692)	$q=32$	2.4699	9.537	10^{-6}	38.05	4.0	DST	2.526	0.26	—	—
256 ² (job 3780692)	$q=64$	2.0808	9.819	10^{-6}	38.05	3.9	DST	2.526	0.26	—	—
256 ² (job 3780692)	$q=128$	1.5497	10.219	10^{-6}	38.05	3.7	DST	2.526	0.25	—	—
256 ² (job 3780692)	$q=256$	0.9689	9.658	10^{-6}	38.05	3.9	DST	2.526	0.26	—	—
256 ² (job 3780692)	linear top rung ($q=R$)	0.7459	3.221	10^{-6}	38.05	11.8	DST	2.526	0.78	—	—
256 ² (job 3780692)	POD $k'=512$	0.1855	10.002	10^{-6}	38.05	3.8	DST	2.526	0.25	—	—
1024 ² (job 3783813)	$q=0$	3.1495	6.394	10^{-6}	103.64	16.2	DST	3.248	0.51	—	—
1024 ² (job 3783813)	$q=32$	2.4641	6.592	10^{-6}	103.64	15.7	DST	3.248	0.49	—	—
1024 ² (job 3783813)	$q=64$	2.0760	6.745	10^{-6}	103.64	15.4	DST	3.248	0.48	—	—
1024 ² (job 3783813)	$q=128$	1.5459	6.875	10^{-6}	103.64	15.1	DST	3.248	0.47	—	—
1024 ² (job 3783813)	$q=256$	0.9648	6.849	10^{-6}	103.64	15.1	DST	3.248	0.47	—	—
1024 ² (job 3783813)	linear top rung ($q=R$)	0.7421	4.424	10^{-6}	103.64	23.4	DST	3.248	0.73	—	—
1024 ² (job 3783813)	POD $k'=512$	0.1838	6.972	10^{-6}	103.64	14.9	DST	3.248	0.47	—	—
64 ² (job 3784662)	head $q=0$ (head_sdf_R512_K16)	4.0451	2.693	10^{-10}	8.25	3.1	SuperLU	1.282	0.48	10.22	3.8
64 ² (job 3784662)	head $q=32$ (head_sdf_R512_K16)	2.7644	2.832	10^{-10}	8.25	2.9	SuperLU	1.282	0.45	10.22	3.6
64 ² (job 3784662)	head $q=64$ (head_sdf_R512_K16)	2.3004	2.846	10^{-10}	8.25	2.9	SuperLU	1.282	0.45	10.22	3.6
64 ² (job 3784662)	head $q=128$ (head_sdf_R512_K16)	2.3580	2.796	10^{-10}	8.25	2.9	SuperLU	1.282	0.46	10.22	3.7
64 ² (job 3784662)	POD $k'=128$	2.5913	2.711	10^{-10}	8.25	3.0	SuperLU	1.282	0.47	10.22	3.8
128 ² (job 3784662)	head $q=0$ (head_sdf_R512_K16)	3.8965	2.673	10^{-10}	14.96	5.6	SuperLU	2.475	0.93	66.51	24.9
128 ² (job 3784662)	head $q=32$ (head_sdf_R512_K16)	2.6261	2.789	10^{-10}	14.96	5.4	SuperLU	2.475	0.89	66.51	23.8
128 ² (job 3784662)	head $q=64$ (head_sdf_R512_K16)	2.1643	2.806	10^{-10}	14.96	5.3	SuperLU	2.475	0.88	66.51	23.7
128 ² (job 3784662)	head $q=128$ (head_sdf_R512_K16)	2.2217	2.868	10^{-10}	14.96	5.2	SuperLU	2.475	0.86	66.51	23.2
128 ² (job 3784662)	POD $k'=128$	2.4833	2.764	10^{-10}	14.96	5.4	SuperLU	2.475	0.90	66.51	24.1
256 ² (job 3784663)	head $q=0$ (head_sdf_R512_K16)	3.8608	2.879	10^{-10}	28.08	9.8	SuperLU	8.386	2.91	484.34	168.2
256 ² (job 3784663)	head $q=32$ (head_sdf_R512_K16)	2.5950	2.963	10^{-10}	28.08	9.5	SuperLU	8.386	2.83	484.34	163.5
256 ² (job 3784663)	head $q=64$ (head_sdf_R512_K16)	2.1305	3.028	10^{-10}	28.08	9.3	SuperLU	8.386	2.77	484.34	159.9
256 ² (job 3784663)	head $q=128$ (head_sdf_R512_K16)	2.2055	3.042	10^{-10}	28.08	9.2	SuperLU	8.386	2.76	484.34	159.2
256 ² (job 3784663)	POD $k'=128$	2.4575	2.850	10^{-10}	28.08	9.9	SuperLU	8.386	2.94	484.34	169.9
512 ² (job 3789568)	head $q=0$ (head_sdf_R512_K16)	3.8521	4.716	10^{-10}	70.12	14.9	SuperLU	36.505	7.74	4672.61	990.8
512 ² (job 3789568)	head $q=32$ (head_sdf_R512_K16)	2.5875	4.812	10^{-10}	70.12	14.6	SuperLU	36.505	7.59	4672.61	971.1
512 ² (job 3789568)	head $q=64$ (head_sdf_R512_K16)	2.1233	4.801	10^{-10}	70.12	14.6	SuperLU	36.505	7.60	4672.61	973.3
512 ² (job 3789568)	head $q=128$ (head_sdf_R512_K16)	2.1977	4.748	10^{-10}	70.12	14.8	SuperLU	36.505	7.69	4672.61	984.1
512 ² (job 3789568)	POD $k'=128$	2.4511	4.031	10^{-10}	70.12	17.4	SuperLU	36.505	9.06	4672.61	1159.2

Table G.5: Burgers at ten times lower viscosity, panel job 3817807 (NVIDIA A100-PCIE-40GB): the neural ladder at fixed $M = 1088$, the $M = 256$ control and the (256, 2176) subject. **F4 applies:** the mesh is under-resolved (the converged discrete operator is 19.3–20.9 % from the 4096² reference), so “vs ref” is the discretisation error and every comparison is reduced-versus-reduced on the same discrete system. Costs are median device ms inside this job; “non-dom.” against all subjects and against reduced subjects only.

subject	worst evolved %	all times %	vs ref %	best-found %	device ms	converged	budget exits	non-dom. (all)	non-dom. (reduced only)
$q = 0, M = 1088$	9.0500	9.0500	17.98	11.3934	557.4	yes	0 no	no	no
$q = 16, M = 1088$	8.7562	8.7562	18.03	11.3462	681.4	yes	0 no	no	no
$q = 32, M = 1088$	8.6360	8.6360	18.08	11.2022	772.2	yes	0 no	no	no
$q = 64, M = 1088$	8.3165	8.3165	18.36	10.9966	913.2	yes	0 no	yes	yes
$q = 128, M = 1088$	7.6276	7.6276	18.70	10.6478	1220.7	yes	0 no	yes	yes
$q = 256, M = 1088$	6.6718	6.6718	19.44	10.0347	1810.3	yes	0 no	yes	yes
$q = 0, M = 256$	8.3933	8.3933	18.00	11.3934	300.3	yes	0 no	no	yes
$q = 16, M = 256$	8.4298	8.4298	17.90	11.3462	379.8	yes	0 no	no	no
$q = 32, M = 256$	8.7191	8.7191	17.75	11.2022	422.7	yes	0 no	no	no
$q = 64, M = 256$	8.9538	8.9538	18.14	10.9966	500.6	yes	0 no	no	no
$q = 128, M = 256$	10.2614	10.2614	19.15	10.6478	658.6	yes	0 no	no	no
$q = 256, M = 2176$	6.1869	6.1869	18.94	10.0347	2864.2	yes	0 no	yes	yes

Table G.6: The same job as Table G.5: POD-LSPG at matched dimension, the free $R = 512$ bank and the full-order tolerance/step grid. **F4 applies** (under-resolved mesh); the non-dominated set against all subjects is nt1e-2_dt01, nt1e-3_dt005, dense_tight, fft_tight.

subject	worst evolved %	all times %	vs ref %	best-found %	device ms	converged	budget exits	non-dom. (all)	non-dom. (reduced only)
POD-LSPG $k' = 16, M = 64$	57.9837	67.3429	60.09	67.3429	46.5	yes	0 no	no	yes
POD-LSPG $k' = 32, M = 128$	45.6825	54.4540	47.84	54.4539	80.2	yes	0 no	no	yes
POD-LSPG $k' = 64, M = 256$	26.2116	35.1340	28.35	35.1339	150.1	yes	0 no	no	yes
POD-LSPG $k' = 128, M = 512$	16.4288	23.2032	23.21	23.2019	331.2	yes	0 no	no	no
POD-LSPG $k' = 256, M = 1024$	11.2540	15.2899	21.96	15.2670	881.2	yes	0 no	no	no
POD-LSPG $k' = 512, M = 2048$	17.7782	17.7782	24.50	10.2797	2735.3	yes	0 no	no	no
free bank $R = 512, M = 1024$	7.7940	7.7940	20.36	7.4455	3174.6	yes	0 no	no	no
nt1e-2_dt01	3.8950	3.8950	19.33	—	14.0	—	— yes	no	no
nt1e-2_dt005	5.2542	5.2542	20.59	—	17.8	—	— no	no	no
nt1e-3_dt01	3.2798	3.2798	19.72	—	63.3	—	— no	no	no
nt1e-3_dt005	0.1971	0.1971	20.86	—	54.9	—	— yes	no	no
nt1e-4_dt01	3.2798	3.2798	19.77	—	88.7	—	— no	no	no
nt1e-4_dt005	0.0571	0.0571	20.81	—	123.0	—	— no	no	no
dense_tight	0.0000	0.0000	20.82	—	112.3	—	— yes	no	no
fft_tight	0.0000	0.0000	20.82	—	171.2	—	— yes	no	no